SPE 25909

Decline-Curve Analysis Using Type Curves--Analysis of Gas Well Production Data

by J.C. Palacio, ECOPETROL(Colombia)/Texas A&M U. and T.A. Blasingame, Texas A&M U.

This is a preprint -- subject to correction.

Copyright 1993, Society of Petroleum Engineers, Inc.

This paper was prepared for presentation at the 1993 SPE Joint Rocky Mountain Regional and Low Permeability Reservoirs Symposium, Denver, CO, April 26-28, 1993.

This paper was selected for presentation by an SPE Program Committee following review of information contained in an abstract submitted by the author(s). Contents of the paper, as presented, have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material, as presented, does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Papers presented at SPE meetings are subject to publication review by Editorial Committees of the Society of Petroleum Engineers. Permission to copy is restricted to an abstract of not more than 300 words. Illustrations may not be copied. The abstract should contain conspicuous acknowledgment of where and by whom the paper is presented. Write Publications Manager, SPE, P.O. Box 833836, Richardson, TX 75083-3836. Telex, 730989 SPEDAL.

ABSTRACT

The analysis of gas production data is presently being done under many constraints that do not allow for rigorous modeling of these data. The traditional efforts not only assume highly idealized production conditions, such as constant bottomhole pressures, but these methods also treat the variation of the gas properties with pressure in an approximate manner at best, and totally neglect these variations at worst.

This paper illustrates theoretical developments and methods of application that can be used to analyze gas well performance data rigorously with decline curve analysis via type curves. These methods are based on the use of "modified time functions" and a new algorithm to compute gas in place which are capable of modeling the behavior of production data for variable rate and/or variable pressure drop conditions. Currently, the only methods to solve for gas in place are iterative schemes. The new algorithm is basically a direct pseudosteady state relation that avoids the necessity of iteration.

In addition, a modified "Fetkovich-Carter" type curve is presented. This new graph gives the performance of constant rate and constant pressure gas flow solutions, the traditional transient flow solutions, and Arps decline curve stems.

Finally, detailed analysis procedures and interpretation strategies are provided. The new analysis techniques introduced are verified with simulation, and the application of the methods are also illustrated using field data.

INTRODUCTION

Decline curve analysis of production data is, in essence, a technique where actual production rate and time data are history matched to a theoretical model using either type curves or computer programs. The theoretical model chosen is then used to predict ultimate oil or gas in place volumes as well as formation properties.

References and illustrations at end of paper

Not only is rate-time analysis founded on proven flow principles but it is also capable of providing virtually the same information as a conventional transient test at no cost in lost production.

The present work introduces new theoretical developments along with their proper implementation techniques that have proven useful in analyzing and interpreting both liquid and gas production with decline curve methods. Emphasis is specially placed on the study of gas flow, and this problem is solved in a "totally liquid equivalent" fashion that allows the liquid solutions to be used to model gas flow. In any case, the previous limitations imposed by either constant bottomhole pressure or constant rate are overridden, making it feasible to analyze production data obtained under actual field conditions.

The new solution for the gas problem is one based on a material balance-like time function and a new algorithm which allow : (1) the use of decline curves developed for liquids, (2) modeling of actual variable rate/variable pressure drop production conditions accurately, and (3) an explicit computation of gas in place.

Another important result of this work is a modified decline type curve that is based on the Fetkovich[1,2] type curve for liquid, and the Carter's[3,4] type curve for gas. This modified curve considers several possibilities of analysis for both single phase gas or liquid and when combined with the rest of the theoretical developments introduced in this work, it can be used to interpret production data regardless of the production scenario.

The tools introduced in this paper are important because (1) they provide a simple, direct and fast way to compute gas in place, (2) they allow inexpensive production tests to replace expensive transient tests, (3) in many cases they avoid the necessity of a long pressure build up especially when dealing with tight, low permeability reservoirs, (4) they are not constrained by highly idealized assumptions regarding the production conditions of either liquid or gas and finally, (5) they rigorously account for the change in gas properties with pressure.

HARMONIC DECLINE FOR LIQUID

In Appendix A of this paper the combination of a material balance relation and the pseudosteady-state flow equation for liquid is shown to yield a very useful theoretical expression for decline curve analysis. This expression is given by Eq. A-10 and is repeated here as Eq. 1.

$$\frac{q_o}{(p_i - p_{wf})}\, b_{pss} = \frac{1}{1 + \left(\frac{m}{b_{pss}}\right)\bar{t}} \quad \text{.....(1)}$$

where

$$m = \frac{1}{Nc_t} \quad \text{.....(2)}$$

$$b_{pss} = \frac{141.2 B_o \mu_o}{k_o h}\left[\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w^2}\right)\right] \quad \text{.....(3)}$$

and

$$\bar{t} = \frac{N_p}{q_o} \quad \text{.....(4)}$$

The dimensionless form of Eq. 1 is

$$q_{Dd} = \frac{1}{1 + \bar{t}_{Dd}} \quad \text{.....(5)}$$

where

$$q_{Dd} = \frac{q_o}{(p_i - p_{wf})}\, b_{pss} \quad \text{.....(6)}$$

and

$$\bar{t}_{Dd} = \left(\frac{m}{b_{pss}}\right)\bar{t} \quad \text{.....(7)}$$

Appendix A proves that Eq. 1 and Eq. 5 should retrace exactly a harmonic decline on a Fetkovich[1,2] type curve. This is done by comparing Eq. 5 to Arps[5] general hyperbolic decline equation also in dimensionless form, given by

$$q_{Dd} = \frac{1}{(1 + b t_{Dd})^{1/b}} \quad \text{.....(8)}$$

Thus, whenever a decline curve exponent, b, of unity is substituted into Eq. 8, the resulting equation has exactly the same form as Eq. 5. This is called a harmonic decline.

This fact implies that as long as the "material balance time", $\bar{t}$, is used rather than ordinary time in the analysis, it is possible to model variable rate and/or variable pressure production scenarios for the single phase liquid case using the Fetkovich[1,2] harmonic (b =1) decline type curve stem.

Eq. 1 is obtained from a relation (Eq. A-5) that is strictly valid for pseudosteady-state. Nevertheless, it has been shown[6,7] that Eq. A-5 works fairly well for transient flow. In other words, when either rate or pressure is changed, a transient is introduced. However, as long as the transient effect remains negligible, compared to the influence of the outer boundary, so that the well is not kept from reaching or maintaining pseudosteady-state like flow, Eq. 1 should yield accurate answers.

The decline curve analysis method proposed here is based on Eqs. 1, 5 and 8. These relations suggest that if $\bar{t}$ is correctly calculated, then a scaled log-log plot of $\frac{q_o}{(p_i - p_{wf})}$ vs. $\bar{t}$ overlays the q_{Dd} versus t_{Dd} trend exactly for a harmonic decline of b=1 on the Fetkovich[1,2] type curve. The relations that should be then used to compute oil in place and reservoir properties are given by Eqs. A-14, A-15 and A-16.

HARMONIC DECLINE FOR GAS

Following a similar procedure presented for liquid, the appropriate gas flow analysis equations are derived in Appendix B. In Appendix B it is shown that for gas the most useful form of the flow equation that should be used to perform decline analysis under variable rate/variable pressure drop conditions is

$$\frac{q_g}{p_{pi} - p_{pwf}}\, b_{a,pss} = \frac{1}{1 + \left(\frac{m_a}{b_{a,pss}}\right)\bar{t}_a} \quad \text{.....(9)}$$

where

$$m_a = \frac{1}{Gc_{ti}} \quad \text{.....(10)}$$

$$b_{a,pss} = 141.2\frac{\mu_{gi}B_{gi}}{k_g h}\left[\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w^2}\right)\right] \quad \text{.....(11)}$$

$$p_p = \frac{\mu_{gi} z_i}{p_i}\int_0^p \frac{p}{\mu_g(p) z(p)}\, dp \quad \text{.....(12a)}$$

and

$$\bar{t}_a = \frac{\mu_{gi} c_{ti}}{q_g}\int_0^t \frac{q_g}{\mu_g(\bar{p}) c_t(\bar{p})}\, dt \quad \text{.....(12b)}$$

The dimensionless form of Eq. 9 is

$$q_{Dd} = \frac{1}{1 + \bar{t}_{a,Dd}} \quad \text{.....(13)}$$

with

$$q_{Dd} = \frac{q_g}{(p_{pi} - p_{pwf})}\, b_{a,pss} \quad \text{.....(14)}$$

and

..... (15)

It must be noted that the q_{Dd} definition is now in terms of normalized pseudopressures and the modified dimensionless decline time function $\bar{t}_{a,Dd}$ is not in terms of real time nor material balance time, but in terms of the material balance pseudotime (Eq. 12b). Still, Eq. 13 should retrace the path of a harmonic decline on the Fetkovich[1,2] type curve because its form is identical to the Arps[5] hyperbolic decline relation (Eq. 8) when a b=1 value is assumed.

This implies that the same techniques of analysis may be employed for both gas and liquids. However, although the computation of the pseudotime function in the gas case involves a much larger effort than the simple material balance time for the liquid case.

Specifically, an estimate of gas in place should be available in order to obtain the correct pseudotime values because that volume would permit the computation of the average reservoir pressure profile and this in turn would allow the estimation of the gas properties involved in the pseudotime integral. However, it is clear that G is an objective rather than an input value for decline curve analysis. Fraim and Wattenbarger[8] propose an iterative scheme on G to solve this problem in their constant bottomhole pressure study. Blasingame and Lee[9] also suggest a similar iterative process in their reservoir limits testing work.

This work introduces a new algorithm that, for the most part, avoids the necessity to iterate on G for the most part. This algorithm is presented in the next section of this paper. Suffice it to say now that as long as the "material balance pseudotime", $\bar{t}_a$, is used rather than ordinary time in the dimensionless decline

function, it is possible to model variable rate and/or variable pressure production scenarios for the single phase gas case, using the Fetkovich[1,2] liquid type curve for a harmonic decline.

Eq. 9 is also derived from a boundary dominated flow relation (Eq. B-4). However, the same comments made for Eq. 1 regarding the validity for both transient and boundary dominated flow are also applicable here.

The decline curve analysis procedure that is proposed for the gas case is based on Eqs. 9, 13 and 8. These relations suggest that if $\bar{t}_a$ is calculated correctly, a scaled log-log plot of $\frac{q_g}{p_{pi} - p_{pwf}}$ vs. $\bar{t}_a$ overlays the original q_{Dd} versus t_{Dd} trend for a harmonic decline of b=1 on the Fetkovich type curve[1,2]. The relations which are used to compute gas in place and reservoir properties are given by Eqs. B-21, B-22 and B-23.

EXPLICIT GAS IN PLACE COMPUTATION

In order to avoid the use of cumbersome iterative techniques to solve the problem at hand, a new procedure to compute gas in place has been devised. The development of the proposed new procedure is presented in Appendix C. The two basic relations that allow that a one-time calculation be made to obtain G with only a few back substitutions are given below. First, we have the material balance equation

$$\frac{1}{G} = \frac{1}{G_p}\left[1 - \frac{\bar{p}}{\bar{z}}\frac{z_i}{p_i}\right] \quad \text{(16)}$$

and second, the gas flow equation for pseudosteady-state flow

$$\bar{p}_p - p_{pwf} = q_g\, b_{a,pss} \quad \text{(17)}$$

The new algorithm makes use of the fact that both G and $b_{a,pss}$ are fixed values in time. Thus, the combination of these two equations permits that the correct $\frac{\bar{p}}{\bar{z}}$ profile in time be obtained. This profile is then used in Eq. 16 and a plot of G vs. time is made. The correct value for G is read from this plot where the computed data trend depicts a horizontal straight line as expected since the original volume in place is not a time dependent value. All this is done in an independent form, making the computation of the average pressure profile much more straightforward yielding a significant simplification for type curve analysis using the new pseudotime function.

After having estimated the correct average reservoir pressure profile using the estimated gas in place based on the extrapolation method given above, the pseudotime values needed for the decline curve analysis can now be computed. This method requires the use of the average pressures to estimate the fluid properties in order to allow integration. The proper estimation of suitable values of these fluid properties is straightforward since we only need to interpolate from a table of PVT data for the particular gas being studied.

MODIFIED FETKOVICH-CARTER TYPE CURVE

This section introduces a new type curve that has been created based on two previous type curves used for decline curve analysis.[1-4] The first of the basic type curves is the one presented by Fetkovich[1,2] which illustrates liquid rigorous solutions and is used to obtain approximate solutions for gas. The second set of curves includes the solution presented by Carter[3,4] which contains rigorous liquid solutions and semi-rigorous gas stems.

The new type curve is referred to simply as the "modified curve" throughout the remainder of this paper.

Type Curve Description

The modified curve is shown here as Fig. 1. The first aspect that needs be mentioned is that the plot combines the original solutions presented by both Fetkovich[1,2] and Carter.[3,4] In order to collect those solutions into a single graph, it was necessary to plot Carter's type curve in terms of Fetkovich equivalent variables. What this means is that since Carter[3,4] presented his plot in terms of simple dimensionless variables (q_D vs. t_D), these variables had to be transformed to their equivalent Fetkovich[1,2] decline dimensionless variables (q_{Dd} vs. t_{Dd}).

The variables used to generate the Fetkovich part of the modified curve are liquid types of variables. On the other hand, Carter's type curve involves pseudopressure in the dimensionless decline rate axis, but still uses actual time for the dimensionless decline time computation. Carter's[3,4] type curve represents a gas solution. However, it must be clear that Carter's[3,4] type curve is not a strictly rigorous technique, because it uses actual time rather than pseudotime, and it assumes a constant bottomhole flowing pressure scenario. This last consideration allows for the variations in gas properties to be accounted for by simply identifying the value of the drawdown parameter, λ. This parameter is the ratio between the viscosity-compressibility product at initial conditions and the value at the average reservoir pressure.

The variables for the type curve generation are given by the following relations which use nomenclature that avoids specifying the type of fluid

$$q_{Dd} = \frac{141.2 B_i \mu_i}{kh} \frac{q}{(p_{fun,i} - p_{fun,wf})}\left[\ln\left(\frac{r_e}{r_w}\right) - \frac{1}{2}\right]$$

$$t_{Dd} = 0.00633 \frac{k}{\phi \mu_i c_{ti} r_w^2} \frac{1}{\frac{1}{2}\left[\frac{r_e^2}{r_w^2} - 1\right]\left[\ln\left(\frac{r_e}{r_w}\right) - \frac{1}{2}\right]} t_{fun}$$

It must be noted that these equations represent a circular reservoir geometry. The p_{fun} variable represents the appropriate pressure function in each case, and t_{fun} is the appropriate time function. For the Fetkovich[1,2] part of the modified curve, p_{fun} is taken as actual pressure and t_{fun} is actual time. The original stems given by Fetkovich[1,2] (liquid case) are the solid lines in the modified curve. On the other hand, for the Carter stems on the type curve, p_{fun} is taken as pseudopressure rather than actual pressure, but t_{fun} is still actual time.

The second important aspect is represented by the additional curves included. There are two additional cases closely related to Carter's[3,4] original type curve. The first of these additions is the inclusion of new λ stems (0.875, 0.625, 0.30) that have values different from the original values (1, 0.75, 0.55). All these stems, original and new, appear as dotted lines on the modified curve.

The next addition is that of the q_{Dd} vs. t_{Dd} stems for different constant rate cases using Carter's strategy. These new solutions are the dashed lines on the modified curve. Carter[3,4] did not include these cases because he was only concerned with the constant bottomhole flowing pressure case. Since Carter's solution is intended to solve gas problems, these new additional curves should prove useful in getting at least an approximate answer when dealing with a gas production scenario under constant flow rate. The generation of these stems was done by means of a t_{Dd} based on actual time and a special q_{Dd} termed q_{Ddc} which is given by

$$q_{Ddc} = \frac{141.2 B_i \mu_i}{kh} \frac{q}{(p_{p,i} - p_{p,abn})}\left[\ln\left(\frac{r_e}{r_w}\right) - \frac{3}{4}\right]$$

where $p_{p,abn}$ is the abandonment pressure of the reservoir. This is done to ensure that these curves indicate production to the point where the reservoir would be totally depleted.

Theory has it that the b=1 solution is the constant rate liquid

solution. Therefore it should be expected that these additional curves lie close to the Fetkovich harmonic stem. It can be seen from the plot that this latter condition is not totally true. The reason why the additional curves do not match the b=1 stem is that gas rather than liquid, is considered. Thus, the fluid property variations are not properly being accounted for since time rather than pseudotime is being used.

The analysis of liquid or gas production data using the modified curve requires suitable definitions of both p_{fun} and t_{fun} according to the needs and desired accuracy. The table given below summarizes the variables that are used for the generation of all the lines on the modified curve.

Table 1 - Generalized Variables for the Modified Type Curve

Type Curve	p_{fun}	t_{fun}
Fetkovich	p	t
Carter (original)	p_p	t
Carter (constant rate)	p_p	t

Type Curve Analysis Methods

We present the two tables below to clarify the usefulness of the different decline curve analysis tools in order to analyze and interpret single phase production data. The tables include the description of the data plotting functions that must be used in each case according to the specific requirements (liquid or gas flow, specific rate and pressure behavior). The additional definitions involved in Table 2 are the normalized pseudopressure and conventional pseudotime which are respectively

$$p_p = \left(\frac{\mu_{gi} z_i}{p_i}\right)\int_0^p \frac{p}{\mu_g z}\, dp$$

and

$$t_a = \mu_{gi} c_{ti} \int_0^t \frac{1}{\bar{\mu}_g \bar{c}_t}\, dt$$

Table 2 - Current Decline Curve Analysis Methods

Case	Pressure Function	Time Function	Method
General Liquid	p	t	Fetkovich b stems as necessary
Liquid constant p_{wf} (rigorous)	p	t	Fetkovich b=0
General Gas (not rigorous)	p_p	t	Fetkovich b stems as necessary
Gas constant p_{wf} (semi-rigorous)	p_p	t	Carter λ stems
Gas constant p_{wf} (rigorous)	p_p	t_a	Fetkovich b=0

In Table 3 the only additional definition required is that of the material balance pseudotime given by Eq. 12. This is the only variable that not only rigorously accounts for the gas properties variation, but also permits any pressure or rate profile for gas flow be studied and matched to a harmonic decline stem.

Table 3 - Decline Curve Analysis Methods Introduced in This Work

Case	Pressure Function	Time function	Method
Liquid, variable pressure/rate (rigorous)	p	$\bar{t} = N_p/q_o$	Fetkovich b=1
Gas, variable pressure/rate (approximation)	p_p	$\bar{t} = G_p/q_g$	Fetkovich b=1
Gas, variable pressure/rate (rigorous)	p_p	$\bar{t}_a$	Fetkovich b=1

ANALYSIS OF GAS PRODUCTION DATA FOR VARIABLE RATE/VARIABLE PRESSURE DROP CONDITIONS

The general analysis procedure provided below includes two options which represent two different forms of estimating the value of gas in place. The first of these options is based on the use of the simple material balance time function, $\bar{t}$, that this work has proven to be theoretically rigorous, yielding a harmonic decline for the case of liquid flow. This approach is proposed because it has been noticed that even though the use of the $\bar{t}$ function is not a rigorous approach for gas, it can still provide an approximate value of gas in place.

The second option makes direct use of the new algorithm described above to estimate the value of gas in place. All the examples in this paper will make use of the new gas in place algorithm so that a full understanding of its capability and functioning can be gained.

For variable pressure drop/variable flow rate conditions the proposed method to analyze gas production data involves the following basic steps:

1. Computation of the Basic Plotting Function for the Vertical Axis

Conversion of pressures (p) to normalized pseudo-pressures (p_p) and computation of the $\frac{q_g}{p_{pi} - p_{pwf}}$ group. This group constitutes the simplest vertical axis plotting function to perform type curve matching.

2. Computation of Gas in Place

Option A :

Calculate the $\bar{t}$ values for each time recorded in the input data set. This function is

$$\bar{t} = \frac{1}{q_g}\int_0^t q_g\, dt = \frac{G_p}{q_g}$$

Compute two complementary plotting functions for the vertical axis based on $\bar{t}$. That is, the integral of $\frac{q_g}{p_{pi} - p_{pwf}}$ with respect to $\bar{t}$ and the derivative of that integral, again with respect to $\bar{t}$. These two additional plotting functions have mathematical forms[10] that follow the well testing practice which intends to keep the units for both the integral and the integral derivative the same as those of the original function. Thus, these plotting functions are given

as :

$$\left(\frac{q_g}{p_{pi} - p_{pwf}}\right)_i = \frac{1}{\bar{t}}\int_0^{\bar{t}} \frac{q_g}{p_{pi} - p_{pwf}}\, d\bar{t}$$

and

$$\left(\frac{q_g}{p_{pi} - p_{pwf}}\right)_{id} = \bar{t}\,\frac{d}{d\bar{t}}\left[\frac{1}{\bar{t}}\int_0^{\bar{t}} \frac{q_g}{p_{pi} - p_{pwf}}\, d\bar{t}\right]$$

The reader is referred to the original references[10] where these definitions and the appropriate type curves were first presented for a thorough discussion on this matter. These variables can be used to perform type curve matching to different decline curves that have been presented previously in the literature.[1,2,10] All the different type curves used here are based on the same principles as the Fetkovich type curve[1,2].

Thus, in all cases, a b=1 stem indicates a harmonic type of decline. Since this case is the fundamental principle on which the new techniques rely, no further comments regarding the type curves will be made and their use will be explained in the example section.

After computing these plotting functions, we match the three different vertical plotting functions versus $\bar{t}$, to liquid decline type curves, such as those given by Fetkovich[1,2] and McCray,[10] in order to estimate the best possible value of gas in place, G.

$$G = \frac{1}{c_{ti}}\frac{(\bar{t})_{M.P.}}{(t_{Dd})_{M.P.}}\frac{\left[\left(\frac{q_g}{(p_{pi} - p_{pwf})}\right)_{fun}\right]_{M.P.}}{(q_{Dd})_{M.P.}}$$

The subscript *"fun"* in the latter relation can be either deleted or substituted with *"i"* or *"id"*. This means that the group with the *"fun"* subscript may be the basic plotting function for the vertical axis, its integral or its integral derivative.

Option B :

Use the new algorithm based on the combination of the material balance equation and the boundary dominated equation for gas flow to obtain the $\bar{p}$ profile. Make a plot of G versus time and determine the best possible value for gas in place, reading it where the line becomes straight and flat. To compute the values of G vs. time, a rearrangement of the material balance equation given by the following relation must be used

$$G = \frac{G_p}{\left(1 - \frac{\bar{p}}{\bar{z}}\frac{z_i}{p_i}\right)}$$

3. Computation of the Rigorous Plotting Function for the Horizontal Axis

Using the G value computed in the previous step, by either method, reevaluate the average reservoir pressure profile in time using the material balance equation, and calculate the material balance pseudotime functions given by Eq. 12. These relations are

$$\frac{\bar{p}}{\bar{z}} = \frac{p_i}{z_i}\left(1 - \frac{G_p}{G}\right)$$

and

$$\bar{t}_a = \frac{\mu_{gi}c_{ti}}{q_g}\int_0^t \frac{q_g}{\mu_g(\bar{p})c_t(\bar{p})}\, dt$$

4. Computation of the Additional Plotting Functions for the Vertical Axis

Compute the complementary plotting functions for the vertical axis based on $\bar{t}_a$. That is, the integral of $\frac{q_g}{p_{pi} - p_{pwf}}$ with respect to $\bar{t}_a$ and the derivative of that integral, again with respect to $\bar{t}_a$. These two additional plotting functions are mathematically[10] expressed in a manner to keep the units for both the integral and the integral derivative the same as those of the original function. In this case, these plotting functions are given[10] as :

$$\left(\frac{q_g}{p_{pi} - p_{pwf}}\right)_i = \frac{1}{\bar{t}_a}\int_0^{\bar{t}_a} \frac{q_g}{p_{pi} - p_{pwf}}\, d\bar{t}_a$$

and

$$\left(\frac{q_g}{p_{pi} - p_{pwf}}\right)_{id} = \bar{t}_a\,\frac{d}{d\bar{t}_a}\left[\frac{1}{\bar{t}_a}\int_0^{\bar{t}_a} \frac{q_g}{p_{pi} - p_{pwf}}\, d\bar{t}_a\right]$$

5. Matching of the Different Plotting Functions to Liquid Type Curves

Use the Fetkovich[1,2] and McCray[10] type curves to match the plotting functions to the harmonic stems of each type curve in all cases, then establish the match point and the appropriate r_{eD} value. Using these latter estimates, compute the formation permeability and skin as well as the final gas in place value. If a circular reservoir geometry is assumed, the following relations should be used :

$$G = \frac{1}{c_{ti}}\frac{(\bar{t}_a)_{M.P.}}{(t_{Dd})_{M.P.}}\frac{\left[\left(\frac{q_g}{(p_{pi} - p_{pwf})}\right)_f\right]_{M.P.}}{(q_{Dd})_{M.P.}}$$

$$k = 141.2\,\frac{B_{gi}\mu_{gi}}{h}\left[\ln\left(\frac{r_e}{r_w}\right) - \frac{1}{2}\right]\frac{\left[\left(\frac{q_g}{(p_{pi} - p_{pwf})}\right)_f\right]_{M.P.}}{(q_{Dd})_{M.P.}}$$

and

$$r_w' = \frac{r_e}{r_{eD}} \qquad s = -\ln\left(\frac{r_w'}{r_w}\right)$$

APPLICATION EXAMPLES

Prior to presenting the examples, we want to emphasize that the design of the type curves allows us to match one curve or set of curves then translate that match to other type curves. Through this arrangement, we can match on the Fetkovich and McCray type curves separately or in unison, then share the results.

Constant Bottom Hole Flowing Pressure Case

The first example is a simulated data set. The pertinent reservoir and fluid data that are input into the simulator are provided in Table 4. A numerical simulator[11] was used to obtain the corresponding rate profile. Then, the step procedure described in the previous section was followed, using option B for the computation of gas in place. The resulting plot of G vs. time is shown here as Fig. 4.

Table 4 - Data Input for First Simulated Example

Reservoir Properties		
Reservoir Area, A	=	40 acres (r_e=745 ft)
Wellbore radius, r_w	=	0.2 ft
Net pay thickness, h	=	30 ft
Porosity, ϕ (fraction)	=	0.3
Permeability, k	=	100 md
Initial Reservoir Pressure, p_i	=	4800 psia
Fluid Properties		
Gas gravity, γ_g	=	0.65 (air=1)
Gas volume factor at p_i, B_{gi}	=	0.6929 RB/Mscf
Gas viscosity at p_i, μ_{gi}	=	2.3485×10^{-2} cp
Gas compressibility at p_i, c_{gi}	=	1.4783×10^{-4} psia^{-1}
Temperature, T	=	200 ° F
Production Characteristics		
Bottomhole pressure, p_{wf}	=	500 psia

It is clear from this plot that the value of 4.15×10^6 Mscf obtained for G with the new method is very close to the actual gas in place (4.04×10^6 Mscf). This verifies, at least partially, the validity of the new algorithm. Using the gas in place value from Fig. 4, the "material balance" pseudotime functions and all the plotting functions described in the previous section are calculated. Figs. 5 and 6 summarize the results in a graphical fashion.

As can be seen from Figs. 5 and 6, a match to a harmonic decline with a b value of unity was always obtained. This example verifies the use of $\bar{t}_q$ for variable rate processes as well as accounting for the variation in gas properties with pressure. The value of the match point value appears as a legend in each plot. Using this match point it is possible to find the final gas in place value, and the formation permeability by means of the equations presented in the preceding section. Thus,

$$G = \frac{1}{(1.4783\times10^{-4}\ \text{psia}^{-1})}\frac{(3.6\ \text{D})}{(1)}\frac{(165\ \text{Mscf/D/psia})}{(1)}$$

$$G = 4.018\times10^{6}\ \text{Mscf}$$

Now, we use a zero value for the skin factor as was considered in the data input into the simulator. This gives

$$k = 141.2\frac{(0.6929\ \text{RB/Mscf})(2.3485\times10^{-2}\ \text{cp})}{30\ \text{ft}}\times \left[\ln\left(\frac{745\ \text{ft}}{0.2\ \text{ft}}\right)-\frac{1}{2}\right]\frac{(165\ \text{Mscf/D/psi})}{(1)}$$

$$k = 97.6\ \text{md}$$

It can be seen that the results obtained after performing the type curve match agree with the input data for both G and k, proving that the analysis is correct.

Variable Flow Rate Case

The second example considers a scenario of step wise changes in gas flow rate and uses the response obtained from inputting such a rate profile into the simulator.[11] The basic input data are the same as those used in the previous example. However the permeability value is changed from 100 md to 1 md. The flow rate is made to vary in step-wise changes and this profile is given by a series of decreasing and increasing rates shown below.

t (days)	q_g (Mscf/D)
0-365	700
366-1095	600
1096-2191	450
2192-2556	500
2557-2922	650
2923-3650	780

The gas in place plot is presented again as both Cartesian and semilog graphs. These graphs are shown here as Figs. 7 and 8. It is clear that the sudden changes in gas flow rate introduce transients in the reservoir. These transients cause some spikes in the expected behavior of G with time, but once a pseudosteady-state flow is reestablished, the data points tend to lie back on the expected horizontal straight line.

It is also seen in Figs. 7 and 8 that the data points corresponding to the last two sudden changes in flow rate do not quite reach a point where they level off to depict a horizontal straight line, as was the case for the first four changes. This behavior can be understood as the result of a transient introduced whose effects did not disappear, preventing the reservoir from returning to full pseudosteady-state flow. Given that the flow relation used in the new algorithm to compute gas in place is certainly a boundary-dominated flow type of equation, the shape of the lines in the gas in place plots is not unexpected.

In spite of this behavior, the adequate horizontal straight line is still readily defined with the first four rate changes, so that the value of gas in place can be estimated with a good degree of accuracy. In fact, in this case the approximate value of gas in place was computed as 4.082×10^6 Mscf which is very close to the actual G value of 4.04×10^6 Mscf.

Now, just as in the previous example, the computed gas in place value is used to calculate the pseudotime functions and then a match to the same type curves used before was attempted. Figs. 9 and 10 show the results obtained for the type curve matching procedure in this case.

Although the variable rate profile produced spikes in the bottomhole flowing pressure values, the magnitude of the pressure changes does not distract from a good estimate of gas in place. The magnified "departure effect" that is caused by an integration of the normalized rate data and later by further differentiation of the results of that integration, is not very significant.

As before it is perfectly reasonable to perform the gas in place and permeability computations using the match point read for any of the three type curves. Using exactly the same procedure and equations employed in the previous example and also assuming that the formation damage parameter, s, has a zero value, the values found for G and k are

$$G = \frac{1}{(1.4783\times10^{-4}\ \text{psi}^{-1})}\frac{(325\ \text{D})}{(1)}\frac{(1.85\ \text{Mscf/D/psi})}{(1)} =$$

$$G = 4.067\times10^{6}\ \text{Mscf}$$

and

$$k = 141.2\frac{(0.6929\ \text{RB/Mscf})(2.3485\times10^{-2}\ \text{cp})}{30\ \text{ft}} \left[\ln\left(\frac{745\ \text{ft}}{0.2\ \text{ft}}\right)-\frac{1}{2}\right]\frac{1.85\ \text{Mscf/D/psia})}{(1)} =$$

$$k = 1.094\ \text{md}$$

The G value from the new algorithm agreed with the actual one of 4.04×10^6 Mscf. The computed k value also agreed with the input value of 1.0 md.

Variable Bottomhole Pressure Case

The third example to be illustrated here is also a simulated case. This case however, involves a production scenario of variable bottomhole flowing pressure. The basic input data are the same as those used in the variable rate example except now the bottomhole pressure is made to vary in step wise changes. The p_{wf} profile is shown below.

t [days]	p_{wf} [psia]
0-365	4000
366-730	2000
731-1095	800
1096-2191	500
2191-3287	350

The gas in place plots obtained in this case are presented as Figs. 11 and 12. The first of these plots is a semilog plot of G vs. time that is presented along with the corresponding Cartesian plot (Fig. 12). These two forms of the G plot are included to help visualize the real effects of the variable pressure profile and also facilitate the choice of the correct G value. In these graphs it is seen that every sudden change in bottomhole flowing pressure introduces a transient in the reservoir that causes some spikes in the expected behavior of G with time. Nevertheless, when pseudosteady-state flow is reestablished, the data points tend to lie back on a horizontal straight line, which is the result expected since G is a fixed value.

The approximate value of G found with the new algorithm is 4.155×10^6 Mscf, which is close to the actual one of 4.04×10^6 Mscf. The former value of G was used to compute the material balance pseudotime functions and then the type curve match was attempted. The results of the type curve matching attempts are shown in Fig. 13. It is clear from this plot that the effect of the transients introduced have a significant effect on the integral curve.

It must be noted in Fig. 13 that the plot of plain normalized rates closely matches Fetkovich harmonic decline. However, the same is not true for the normalized rate integral.

This departure from the correct model is due to the fact that even though the sudden changes in bottomhole flowing pressures still allow us to use the "raw" rate data, when these data are integrated the effect is magnified making it very difficult to model the situation with boundary dominated flow equations. In other words, the simulator response to a sudden reduction in p_{wf} always includes flow rate values that are far from being feasible in any actual production field (the rate spikes usually diminish in less than two days). When these "spike" rates are used, as such, for type curve matching the boundary dominated flow equations are still capable of modeling the situation fairly accurately. However, the spike rates make it difficult to match these data to the appropriate integral type curve.

In spite of the previous considerations, Fig. 13 shows that although the data do not match McCray's type curve, a trend is still observed that seems to approach the harmonic decline stem.

Using the normalized rate data does yield a good match to Fetkovich type curve, the required values of G and k were then computed in exactly the same form used in the previous example.

$$G = \frac{1}{(1.4783 \times 10^{-4}\ \text{psia}^{-1})} \frac{(320\ \text{D})}{(1)} \frac{1.9\ \text{Mscf/D/psia})}{(1)}$$

$$G = 4.113 \times 10^6\ \text{Mscf}$$

and

$$k = 141.2 \frac{(0.6929\ \text{RB/Mscf})(2.3485 \times 10^{-2}\ \text{cp})}{30\ \text{ft}} \left[\ln\left(\frac{745\ \text{ft}}{0.2\ \text{ft}}\right) - \frac{1}{2}\right] \frac{1.9\ \text{Mscf/D/psia})}{(1)}$$

$$k = 1.12\ \text{md}$$

The value obtained for G from type curve matching compares very well with both the value obtained from the algorithm and the actual one of 4.04×10^6 Mscf. The same good agreement is seen for the calculated k as compared to the actual value of 1.0 md.

Field Case Example

As a field example we present Gas Well A which is in a low permeability gas reservoir. That is, the permeability, k, is generally less than 0.1 md. The bottomhole flowing pressures and flow rates are recorded on a daily basis. Finally, very significant liquid loading and seasonal effects have been detected.

The following table summarizes the values of the pertinent formation and fluid properties.

Table 5 - Data Input for Field Example

Reservoir Properties		
Reservoir Area, A	=	6.6 acres (r_e = 303 ft)
Wellbore radius, r_w	=	0.3 ft
Net pay thickness, h	=	48 ft
Porosity, ϕ (fraction)	=	0.09
Initial Reservoir Pressure, p_i	=	2625 psia
Fluid Properties		
Gas gravity, γ_g	=	0.72 (air = 1)
Gas volume factor at p_i, B_{gi}	=	1.166 RB/Mscf
Gas viscosity at p_i, μ_{gi}	=	$1.785 * 10^{-2}$ cp
Gas compressibility at p_i, c_{gi}	=	$3.649 * 10^{-4}$ psia^{-1}
Temperature, T	=	275 ° F

The type of variation in bottomhole flowing pressures and flow rates present in the well are shown in Figs. 14 and 15. It is clear from Fig. 14 that p_{wf} varies widely with no defined patterns.

Fig. 15 is a logarithmic plot of normalized instantaneous rates versus real time which shows that a match of this data set to any decline type curve is impossible at this point. In addition, Fig. 16 shows a poor performance for pressure drop normalized rate versus time.

The proposed procedure to analyze this kind of data was then used once more with option "B", and the resulting gas in place plots are shown here as Figs. 17 and 18. It can be seen that these plots do not exhibit the horizontal straight line that should be obtained for the fixed value of G. In fact, the poor performance observed was not unexpected given the production characteristics of this well. In particular, it must be remembered that all the theory presented in this work assumes single phase gas flow, and the significant liquid loading that exists in this well may invalidate this fundamental assumption.

Although the gas in place plots are far from ideal, a tendency toward a flat region and a high concentration of points is seen around a G of 0.2×10^6 Mscf. In order to determine if an interpretation of the data set from Gas Well A is possible at all by our method, this G value is used to compute the material balance pseudotimes. After doing so, the normalized rate profile is replotted versus the calculated material balance pseudotimes. This new plot is presented here as Fig. 19.

The comparison of Figs. 16 and 19 shows a significant

improvement in smoothness in the latter one, so that a match of the normalized rate profile of this well can now be attempted. The results of the type curve matching procedures are shown in Figs. 20 and 21. In this case when Fetkovich and McCray's type curves are used, the well production data are found to lie on a b=1 stem, indicating a harmonic decline as predicted from theory.

The agreement with the theoretical principles presented in this work is a very important result because of the unfavorable conditions involved in the analysis of this case. The match to the harmonic stems allows the computation of a corrected gas in place volume. Thus, the value of the gas in place is

$$G = \frac{1}{(3.649\text{x}10^{-4}\ \text{psia}^{-1})}\frac{(115\ \text{D})}{(1)}\frac{(0.78\ \text{Mscf/D/psia})}{(1)}$$

$$G = 0.245\text{x}10^{6}\ \text{Mscf}$$

and also

$$k = 141.2\frac{(1.166\ \text{RB/Mscf})(1.785\text{x}10^{-2}\ \text{cp})}{48\ \text{ft}}\text{x}$$
$$\left[\ln\left(\frac{303\ \text{ft}}{0.3\ \text{ft}}\right) - \frac{1}{2}\right]\frac{(0.78\ \text{Mscf/D/psia})}{(1)}$$

$$k = 0.306\ \text{md}$$

This corrected gas in place of 0.245x10^6 Mscf is not unreasonable compared to the value provided by the algorithm. Also, any value of permeability in the area is expected to be within tenths of millidarcies, so the value of 0.306 md represents a reasonable estimate for this formation parameter. The knowledge of the skin factor would improve the estimate of the permeability.

These conclusions make it possible to state that even for non-ideal situations, the explicit calculation of gas in place with the new algorithm provides an accurate result. Furthermore, the calculated gas in place permits the computation of the pseudotimes which are required for analysis of gas production data under actual field conditions, the match to a harmonic stem and the later computation of the permeability value.

SUMMARY AND CONCLUSIONS

SUMMARY

This work has introduced new methods of production data analysis for the case of single phase flow of either oil or gas using type curve analysis. The gas flow techniques have been presented in a "liquid equivalent form". This ensures that the techniques for analysis, or at least their form, are the same for either type of fluid.

CONCLUSIONS

1. The use of a material balance time function for the case of single phase liquid production allows for the analysis of any data set using the b=1 stem on Fetkovich type curve regardless of the rate or pressure conditions under which those data were obtained. This modified time function must be taken as the quotient between cumulative liquid production and flow rate.
2. The use of a modified definition of the conventional pseudotime as proposed by Agarwal[12] that resembles a material balance type of function permits a pseudosteady-state gas flow equation to be obtained which has the same form as that for the liquid case. This modified definition has previously been proven accurate by simulation techniques , and now has been proven to be analytically correct and have an exact theoretical definition.
3. An accurate technique has been devised that allows for the explicit computation of gas in place from field data. This technique makes the analysis of gas data using the modified pseudotime function much simpler.
4. The harmonic type of decline (b=1) is the "all-inclusive" solution because it is useful for any sort of analysis regardless of the fluid type or the conditions under which the data are obtained. The only requirement for the validity of this statement is that the proper plotting variables be used in each case, so that a match can be found.
5. The analysis and interpretation of production data are relatively easy and can provide just about the same information as conventional pressure transient analysis. Therefore production data analysis should be used in reservoir engineering because such analysis would save time and money for it involves no cost in lost production.
6. The flow equations derived in this paper for both liquid and gas which lead to harmonic declines are strictly valid for boundary dominated flow. However, they constitute reasonable approximations for transient flow conditions.
7. The flow rate integral and flow rate integral derivative functions are helpful to obtain a more accurate match to decline type curves than using flow rate data alone. This is true for production conditions as those actually found in the field.

NOMENCLATURE

Dimensionless and Field Variables

A = Drainage Area, ft^2
b = Fetkovich/Arps Decline Curve Exponent
b_{pss} = Constant in the Liquid Flow Equation Defined by Eqs. A-9 or 3
$b_{a,pss}$ = Constant in the Gas Flow Equation Defined by Eqs. B-8 or 11
B_g = Gas Formation Volume Factor, RB/Mscf
B_{gi} = Gas Formation Volume Factor at Initial Reservoir Pressure, RB/Mscf
B_o = Oil Formation Volume Factor, RB/STB
C_A = Reservoir Shape Factor
c_g = Gas Compressibility, psia^{-1}
$\bar{c}_g$ = Gas Compressibility at Average Reservoir Pressure, psia^{-1}
c_{gi} = Gas Compressibility at Original Reservoir Pressure, psia^{-1}
c_t = Total System Compressibility, psia^{-1}
c_{ti} = Total System Compressibility at Original Reservoir Pressure, psia^{-1}
G = Original Gas in Place, Mscf
G_p = Cumulative Gas Production, Mscf
h = Formation Thickness, ft
k_o = Effective Permeability to Oil, md
k_g = Effective Permeability to Gas, md
m = Constant Defined by Eq. A-8 or 2
m_a = Constant Defined by Eq. B-7 or 10
M = Gas Molecular Weight
N = Original Oil in Place, STB
N_p = Cumulative Oil Production, STB
p = Pressure, psia
$\bar{p}$ = Average Reservoir Pressure, psia
p_{fun} = Appropriate Pressure-like Type of Function for Every Stem in the Modified Type Curve, psia
p_i = Initial Reservoir Pressure, psia
p_p = Normalized Pseudopressure, psia

$\bar{p}_p$ = Normalized Pseudopressure at Average Reservoir Pressure, psia
$p_{p,abn}$ = Normalized Pseudopressure at Abandonment Pressure, psia
p_{pi} = Normalized Pseudopressure at Initial Reservoir Pressure, psia
p_{pwf} = Normalized Pseudopressure at Wellbore Flowing Pressure, psia
p_{wf} = Wellbore Flowing Pressure, psia
q = Surface Flow Rate, STB/D or Mscf/D
q_D = Constant Pressure Dimensionless Rate Solution
q_{Dd} = Dimensionless Decline Rate Function
q_{Ddc} = Dimensionless Decline Rate Function Used to Generate Carter's Type Curves for Constant Rate
q_{Ddi} = Dimensionless Decline Rate Integral
q_{Ddid} = Dimensionless Decline Rate Integral Derivative
q_i = Initial Rate for Oil or Gas, STB/D or Mscf/D
q_g = Gas Flow Rate, Mscf/D
q_o = Oil Flow Rate, STB/D
r_e = Drainage Radius, ft
r_{eD} = Dimensionless Drainage Radius
r_w = Wellbore Radius, ft
r_w' = Effective Wellbore Radius (Includes Formation Damage Effects), ft
R = Constant for Real Gases in the Equation of State
s = Formation Damage (Skin) Factor
t = Time, days
$\bar{t}$ = Material Balance Time (N_p/q_o or G_p/q_g), days
t_a = Conventional Normalized Pseudotime, days
$\bar{t}_a$ = Normalized Material Balance Pseudotime, days
$t_{a,DA}$ = Dimensionless Time Based on Drainage Area and Normalized Conventional Pseudotime
$\bar{t}_{a,DA}$ = Dimensionless Time Based on Drainage Area and Normalized Material Balance Pseudotime
$\bar{t}_{a,Dd}$ = Dimensionless Decline Time Based on Normalized Material Balance Pseudotime
t_D = Dimensionless Time Based on Wellbore Radius
t_{DA} = Dimensionless Time Based on Drainage Area
$\bar{t}_{DA}$ = Dimensionless Decline Time Based on Drainage Area and Material Balance Time
t_{Dd} = Dimensionless Decline Time
$\bar{t}_{Dd}$ = Dimensionless Decline Time Based on Material Balance Time
t_{fun} = Appropriate Time-like Type of Function for Every Stem in the Modified Type Curve, days
t_{pss} = Time for the Onset of Steady State, days
T = Temperature, °F
z = Real Gas Deviation Factor
$\bar{z}$ = Real Gas Deviation Factor at Average Reservoir Pressure
z_i = Real Gas Deviation Factor, at Initial Reservoir Pressure

Greek Variables

ϕ = Formation Porosity, fraction
γ = Euler's Constant = 0.577216....
γ_g = Gas Specific Gravity
λ = Carter's Drawdown Parameter
μ = Fluid Viscosity, cp
μ_g = Gas Viscosity, cp
$\bar{\mu}_g$ = Gas Viscosity at Average Reservoir Pressure, cp
μ_{gi} = Gas Viscosity at Initial Reservoir Pressure, cp
μ_o = Oil Viscosity, cp
ρ = Gas Density, lb/mol

Special Subscripts

a = "adjusted" variable for gas well test analysis. Use of these variables in gas well test analysis yields an equivalent liquid system.
D = dimensionless variable
$M.P.$ = Match Point
pss = Pseudosteady-state

REFERENCES

1. Fetkovich, M.J.: "Decline Curve Analysis Using Type Curves," *JPT* (June 1980) 1065-77.
2. Fetkovich, M.J., *et al*: "Decline-Curve Analysis Using Type Curves--Case Histories," *SPEFE* (Dec. 1987) 637-56.
3. Carter, R.D.: "Characteristic Behavior of Finite Radial and Linear Gas Flow Systems - Constant Terminal Pressure Case," paper SPE 9887 presented at the 1981 SPE/DOE Low Permeability Symposium, Denver, Colorado, May 27-29.
4. Carter, R.D.: "Type Curves for Finite Radial and linear Gas Flow Systems: Constant Terminal Pressure Case," *SPEJ* (Oct. 1985) 719-728.
5. Arps, J.J.: "Analysis of Decline Curves," ***Trans.*** AIME (1945) 160, 228-247.
6. Blasingame, T.A. and Lee, W.J.: "The Variable-Rate Reservoir Limits Testing," paper SPE 15028 presented at the SPE Permian basin Oil & Gas Recovery Conference, Midland, March 13-14, 1986.
7. Blasingame, T.A., McCray, T.C. and Lee, W.J.: "Decline Curve Analysis for Variable Pressure Drop/Variable Flowrate Systems," paper SPE 21513 presented at the SPE Gas Technology Symposium, Houston, June 23-24, 1991.
8. Fraim, M.L. and Wattenbarger, R.A.: "Gas Reservoir Decline-Curve Analysis Using Type Curves With Real Gas Pseudopressure and Normalized Time," *SPEFE* (Dec. 1987) 671-82.
9. Blasingame, T.A- and Lee, W.J.: "The Variable-Rate Reservoir Limits Testing of Gas Wells," paper SPE 17708 presented at the 1988 SPE Gas Technology Symposium, Dallas, TX, June 13-15, 1988.
10. McCray, T.L.: *Reservoir Analysis Using Production Decline Data and Adjusted Time*, M.S. Thesis, Texas A&M University, College Station, TX (1990).
11. Holditch, S.A., and Associates: "Sabre - A General Purpose Three-Dimensional, Three-Phase Petroleum Reservoir Simulator - User's Manual", College Station, TX, May, 1990.
12. Agarwal, R.G.: "Real Gas Pseudo Time - A New Function for Pressure Buildup Analysis of MHF Gas Wells," paper SPE 8729 presented at the 1979 SPE Annual technical conference and exhibition, Las Vegas, Sept. 23-26.
13. Dake, L.P. and Hurst, W.: "*Fundamentals of Reservoir Engineering*," Elsevier Scientific Publishing Company , 1978, 145-150.
14. Meunier, D.F., Kabir, C.S., Wittmann, M.J.: "Gas Well Test Analysis : Use of Normalized Pseudovariables," *SPEFE* (Dec. 1987) 629-636.
15. Al-Hussainy, R., Ramey, H.J. Jr. and Crawford, P.B. : "The Flow of Real Gases through Porous Media," *JPT* (May 1966) 624-636; ***Trans.*** AIME, **237**.
16. Al-Hussainy, R. and Ramey, H.J. Jr. : "Application of Real

Gas Flow Theory to Well Testing and Deliverability Forecasting," *JPT* (May 1966) 637-642; ***Trans.*** AIME, **237.**

APPENDIX A - DEVELOPMENT AND USE OF THE MATERIAL BALANCE TIME FOR BOUNDARY DOMINATED LIQUID FLOW

DEVELOPMENT

The original material balance approach for boundary dominated flow (i.e., pseudosteady-state) was developed by Blasingame and Lee,[6] and the following derivation builds up on their developments.

From the compressibility definition it can be shown that

$$q_o = -\frac{A\phi h\, c_t}{5.615\, B_o}\frac{d\bar{p}}{dt} \quad \text{.....(A-1)}$$

Since the single phase liquid case is the one under consideration, c_t is assumed to be constant. Thus, the integration of Eq. A-1 produces

$$\int_0^t q_o\, dt = -\frac{A\phi h\, c_t}{5.615\, B_o}\int_{p_i}^{\bar{p}} d\bar{p}$$

Also, since the integral on the left hand side corresponds to the cumulative production,

$$N_p = \frac{A\phi h\, c_t}{5.615\, B_o}(p_i - \bar{p})$$

Solving this relation for the pressure difference involved, it follows that

$$(p_i - \bar{p}) = \frac{5.615\, N_p\, B_o}{\phi h A c_t} = \frac{N_p}{N\, c_t} \quad \text{.....(A-2)}$$

which may be also rearranged as

$$(p_i - \bar{p})\frac{h}{B_o} = \frac{5.615\, N_p}{\phi A c_t}$$

and further rewritten as

$$(p_i - \bar{p})\frac{k_o h}{141.2 q_o B_o \mu_o} = \frac{5.615\, k_o}{141.2 \phi \mu_o c_t A}\frac{N_p}{q_o}$$

Now, letting $\bar{t} = \frac{N_p}{q_o}$ be the material balance time function it follows that

$$(p_i - \bar{p})\frac{k_o h}{141.2 q_o B_o \mu_o} = \frac{2\pi\,(0.00633)\, k_o}{\phi \mu_o c_t A}\bar{t}$$

Furthermore, if the dimensionless time is redefined as

$$\bar{t}_{DA} = \frac{0.00633 k_o}{\phi \mu_o c_t A}\bar{t} \quad \text{.....(A-3)}$$

then, the next to last relation becomes

$$(p_i - \bar{p})\frac{k_o h}{141.2 q_o B_o \mu_o} = 2\pi \bar{t}_{DA} \quad \text{.....(A-4)}$$

The most important characteristic of Eq. A-4 is that it is always valid regardless of time, flow regime, or production scenario whether it is constant or variable bottomhole flowing pressure or constant or variable flow rate. This is due to the fact that Eq. A-4 is a material balance type of equation.

Now, it has been proved[13] that for the scheme of constant rate production of single phase liquid the flow equation for the pressure response under boundary dominated flow may be written as

$$(\bar{p} - p_{wf})\frac{k_o h}{141.2 q_o B_o \mu_o} = \frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w'^2}\right) \quad \text{.....(A-5)}$$

Although Eq. A-5 was derived for constant rate (variable p_{wf}), it has been shown[6] in the literature to be a good approximation when the flowing bottomhole pressure is fixed (variable rate).

Thus, the addition of Eqs. A-4 and A-5 yields

$$(p_i - p_{wf})\frac{k_o h}{141.2\, q_o B_o \mu_o} = 2\pi \bar{t}_{DA} + \frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w'^2}\right) \quad \text{...(A-6)}$$

The previous considerations imply that Eq. A-6 is strictly valid for the pseudosteady-state flow regime and any rate or pressure profile. If Eqs. A-3 and A-6 are combined and the result rearranged, then

$$\frac{p_i - p_{wf}}{q_o} = m\bar{t} + b_{pss} \quad \text{.....(A-7)}$$

where

$$m = \frac{2\pi(0.00633)k_o}{\phi \mu_o c_t A}\frac{141.2\, B_o \mu_o}{k_o h} = \frac{5.615\, B_o}{\phi h c_t A} = \frac{1}{N c_t} \quad \text{...(A-8)}$$

and

$$b_{pss} = \frac{141.2 B_o \mu_o}{k_o h}\left[\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w'^2}\right)\right] \quad \text{.....(A-9)}$$

or finally,

$$\frac{q_o}{(p_i - p_{wf})}\, b_{pss} = \frac{1}{1 + \left(\frac{m}{b_{pss}}\right)\bar{t}} \quad \text{..... (A-10)}$$

USE

The group on the L.H.S. of Eq. A-10 is exactly the dimensionless decline rate variable, q_{Dd}, as presented by Fetkovich[1,2]. The second term in the denominator of the R.H.S. group in the same equation is defined as, $\bar{t}_{Dd}$, therefore

$$\bar{t}_{Dd} = \left(\frac{m}{b_{pss}}\right)\bar{t} = \frac{2\pi\, \bar{t}_{DA}}{\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w'^2}\right)} \quad \text{..... (A-11)}$$

The only difference between $\bar{t}_{Dd}$ and the Fetkovich[1,2] dimensionless decline time function, t_{Dd}, is that now real time t is substituted with $\bar{t}$. Since Fetkovich[1,2] assumed a circular reservoir, the denominator on the R.H.S. of Eq. A-11 is given by $\ln\left(\frac{r_e}{r_w} - \frac{3}{4}\right)$ which is the result of taking the appropriate C_A value for a circular reservoir. In addition Fetkovich[1,2] actually chose to use 1/2 rather than 3/4 within the argument of the logarithm, because he found a better correlation in doing so.

In any case, Eq. A-10 can be reduced to

$$q_{Dd} = \frac{1}{1 + \bar{t}_{Dd}} \quad \text{..... (A-12)}$$

Eq. A-12 is a harmonic decline type of equation where the decline exponent is unity as required. This is evident by comparing Eq. A-12 to Arps[5] original definition of a hyperbolic decline which in dimensionless form is

$$q_{Dd} = \frac{1}{(1 + b t_{Dd})^{1/b}} \quad \text{..... (A-13)}$$

When the decline exponent, b, is taken as unity in Arps[5] equation,

a harmonic decline is obtained and Eqs. A-12 and A-13 would have exactly the same form.

Therefore if $\bar{t}$ is correctly calculated, a scaled log-log plot of $\frac{q_o}{(p_i - p_{wf})}$ versus $\bar{t}$ will overlay the q_{Dd} versus $\bar{t}_{Dd}$ trend for a harmonic decline on Fetkovich's[1,2] type curve. Once the match has been obtained, the relations to get m, b_{pss}, and N are

$$b_{pss} = \frac{(q_{Dd})_{M.P.}}{\left(\frac{q_o}{(p_i - p_{wf})}\right)_{M.P.}} \quad \text{(A-14)}$$

$$m = \frac{1}{Nc_{ti}} = b_{pss}\frac{(t_{Dd})_{M.P.}}{(\bar{t})_{M.P.}} \quad \text{(A-15)}$$

$$N = \frac{1}{c_{ti}}\frac{(\bar{t})_{M.P.}}{(t_{Dd})_{M.P.}}\frac{\left(\frac{q_o}{(p_i - p_{wf})}\right)_{M.P.}}{(q_{Dd})_{M.P.}} \quad \text{(A-16)}$$

where *M.P.* stands for a match point value.

APPENDIX B - DEVELOPMENT, PROOF AND USE OF THE MATERIAL BALANCE PSEUDO-TIME FOR BOUNDARY DOMINATED GAS FLOW

DEVELOPMENT

Blasingame and Lee[9] proposed a modified gas flow equation based on modified pseudotime function to model the general variable rate/variable pressure drop case. This was done on a empirical basis using simulation to verify this approach. Following are the analytical proofs that constitute the sound theoretical background for Blasingame and Lee's[9] modified gas flow equation.

Thus, taking the material balance equation for liquid given by

$$\frac{(p_i - \bar{p})}{q_o} = \frac{1}{N\,c_t}\frac{N_p}{q_o} = \frac{1}{N\,c_t}\bar{t} \quad \text{(B-1)}$$

as the basis for their direction of thought, Blasingame and Lee[9] proposed that, in the gas case

$$\frac{(p_{pi} - \bar{p}_p)}{q_g} = \frac{1}{G\,c_t}\bar{t}_a \quad \text{(B-2)}$$

is valid when the new pseudotime variable is a material balance type of relation given by

$$\bar{t}_a = \frac{\mu_{gi}c_{ti}}{q_g}\int_0^t \frac{q_g}{\mu_g(\bar{p})c_t(\bar{p})}\,dt \quad \text{(B-3)}$$

The pseudopressure functions involved in Eq. B-2 are normalized types of variables, already published in the literature[14] which are given by

$$p_{pi} = \left(\frac{\mu_{gi}\,z_i}{p_i}\right)\int_0^{p_i}\frac{p}{\mu_g z}\,dp$$

and

$$\bar{p}_p = \left(\frac{\mu_{gi}\,z_i}{p_i}\right)\int_0^{\bar{p}}\frac{p}{\mu_g z}\,d\bar{p}$$

Eq. B-2 was coupled with the pseudosteady-state flow equation given by Al Hussainy and Ramey[15,16] for the flow of single phase gas, which is

$$\frac{(p_{pi} - \bar{p}_p)}{q_g} = 141.2\frac{\mu_{gi}B_{gi}}{k_g h}\left[\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w'^2}\right)\right] \quad \text{(B-4)}$$

Thus, the addition of Eq. B-3 and B-4 produced the following new gas equation for variable rate post transient flow

$$\frac{\Delta p_p}{q_g} = m_a\bar{t}_a + b_{a,pss} \quad \text{(B-5)}$$

where

$$\Delta p_p = (p_{pi} - p_{pwf}) \quad \text{(B-6)}$$

$$m_a = \frac{1}{Gc_{ti}} \quad \text{(B-7)}$$

$$b_{a,pss} = 141.2\frac{\mu_{gi}B_{gi}}{k_g h}\left[\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w'^2}\right)\right] \quad \text{(B-8)}$$

PROOF

The gas compressibility definition is

$$c_g = \frac{1}{\rho}\frac{d\rho}{dp} = \frac{zRT}{pM}\frac{d}{dp}\left(\frac{pM}{zRT}\right) \quad \text{(B-9)}$$

Evaluating this definition at the average reservoir pressure and eliminating constant terms, it follows that

$$\frac{d}{d\bar{p}}\left(\frac{\bar{p}}{\bar{z}}\right) = \frac{\bar{p}}{\bar{z}}\bar{c}_g \quad \text{(B-10)}$$

Also, solving the gas material balance equation for cumulative production and differentiating the resulting relation, it is readily seen that

$$q_g = -\frac{G\,z_i}{p_i}\frac{d}{dt}\left(\frac{\bar{p}}{\bar{z}}\right) \quad \text{(B-11)}$$

Now, if the chain rule is used on Eq. B-11, it follows that

$$q_g = \frac{dG_p}{d\bar{p}} = -\frac{G\,z_i}{p_i}\frac{d}{d\bar{p}}\left(\frac{\bar{p}}{\bar{z}}\right)\frac{d\bar{p}}{dt} \quad \text{(B-12)}$$

which, combined with Eq. B-10, produces

$$q_g = -\frac{G\,z_i}{p_i}\frac{\bar{p}}{\bar{z}}\bar{c}_g\frac{d\bar{p}}{dt} \quad \text{(B-13)}$$

If Eq. B-13 is substituted into Eq. B-3, it is seen that

$$\bar{t}_a = \frac{\mu_{gi}\,c_{ti}}{q_g}\left(-\frac{G\,z_i}{p_i}\right)\int_0^t \frac{\bar{p}}{\bar{z}}\bar{c}_g\frac{1}{\bar{\mu}_g\,\bar{c}_t}\frac{dp}{dt}\,dt \quad \text{(B-14)}$$

Assuming that the formation compressibility is negligible compared to the fluid compressibility, then $\bar{c}_t \cong \bar{c}_g$. Thus, using this approximation and changing the variable of integration from time to pressure, Eq. B-14 may be written as

$$\bar{t}_a = -\frac{G\,c_{ti}}{q_g}\left(\frac{\mu_{gi}\,z_i}{p_i}\right)\int_{p_i}^{\bar{p}}\frac{p}{\mu_g z}\,dp \quad \text{(B-15)}$$

Furthermore, using the definition of normalized pseudopressure, Eq. B-15 may be represented by

$$\bar{t}_a = G\,c_{ti}\frac{p_{pi} - \bar{p}_p}{q_g}$$

which can be rearranged to yield

$$\frac{(p_{pi} - \bar{p}_p)}{q_g} = \frac{1}{G\,c_t}\bar{t}_a \quad \text{(B-16)}$$

Eq. B-16 is identical to Eq. B-2. This fact constitutes the proof that Eq. B-3 is the exact definition of pseudotime for boundary dominated gas flow.

USE

The use of the tools developed above for decline curve analysis lies in the fact that Eq. B-5 can undergo exactly the same rearrangement process used for Eq. A-7. Thus, Eq. B-5 can be rewritten as

$$\frac{q_g}{\Delta p_p} b_{a,pss} = \frac{1}{1 + \left(\frac{m_a}{b_{a,pss}}\right)\bar{t}_a} \quad \text{(B-17)}$$

Eq. B-17 and the liquid relation given by Eq. A-10 have identical forms and the former can be rewritten as

$$q_{Dd} = \frac{1}{1 + \bar{t}_{a,Dd}} \quad \text{(B-18)}$$

where now

$$q_{Dd} = \frac{q_g}{\Delta p_p} b_{a,pss} \quad \text{(B-19)}$$

and

$$\bar{t}_{a,Dd} = \left(\frac{m_a}{b_{a,pss}}\right)\bar{t}_a \quad \text{(B-20)}$$

Literally, the dimensionless decline rate function is now in terms of pseudopressure rather than actual pressure and the dimensionless decline time is in terms of the material balance pseudotime. Eq. B-18 and the relation for liquid given by Eq. A-12 are identical in form, as is expected. Therefore, all of the comments regarding the comparison of Eq. A-12 to Arps[5] relations (i.e., a harmonic decline) apply here in the same fashion.

This indicates that when pseudopressure and the material balance pseudotime are used to model gas flow, the data trend must decline along the harmonic stem on Fetkovich[1,2] liquid type curve. That is, if $\bar{t}_a$ is correctly calculated then a scaled log-log plot of $\frac{q_g}{(p_{pi} - p_{pwf})}$ versus $\bar{t}_a$ will overlay the q_{Dd} versus $\bar{t}_{Dd}$ trend exactly for a b=1 stem on Fetkovich's[1,2] type curve. The match point permits computation of the gas in place and formation properties from the following relations

$$b_{a,pss} = \frac{(q_{Dd})_{M.P.}}{\left(\frac{q_g}{(p_{pi} - p_{pwf})}\right)_{M.P.}} \quad \text{(B-21)}$$

$$m_a = \frac{1}{Gc_{ti}} = b_{a,pss}\frac{(t_{Dd})_{M.P.}}{(\bar{t}_a)_{M.P.}} \quad \text{(B-22)}$$

$$G = \frac{1}{c_{ti}}\frac{(\bar{t})_{M.P.}}{(t_{Dd})_{M.P.}}\frac{\left(\frac{q_g}{(p_{pi} - p_{pwf})}\right)_{M.P.}}{(q_{Dd})_{M.P.}} \quad \text{(B-23)}$$

where $M.P.$ stands for "Match Point". The last three relations form the basis for analysis when this harmonic type curve technique is used.

APPENDIX C - AN EXPLICIT METHOD FOR COMPUTING GAS-IN-PLACE--AN AID FOR THE COMPUTATION OF PSEUDOTIME

In order to avoid the use of iterative schemes to compute the gas in place, G, a new technique is presented to couple the gas material balance equation and the pseudosteady state flow relation. The final result of this combination is an algorithm that needs be solved only once to estimate the value of gas in place. Solving the material balance equation for a volumetric dry gas reservoir for G, it follows that

$$\frac{1}{G} = \frac{1}{G_p}\left[1 - \frac{\bar{p}}{\bar{z}}\frac{z_i}{p_i}\right] \quad \text{(C-1)}$$

Since G is a fixed value, Eq. C-1 can be written for two different points in time; that is, for two different values of average reservoir pressure. Thus, for times j and $j+1$, Eq. C-1 can be arranged to yield

$$\frac{1}{G_{p,j+1}}\left[1 - \frac{\bar{p}_{j+1}}{\bar{z}_{j+1}}\frac{z_i}{p_i}\right] = \frac{1}{G_{p,j}}\left[1 - \frac{\bar{p}_j}{\bar{z}_j}\frac{z_i}{p_i}\right] \quad \text{(C-2)}$$

Eq. C-2 may be then rearranged as follows

$$\left(\frac{\bar{p}}{\bar{z}}\right)_j = \frac{p_i}{z_i}\left\{1 - \frac{G_{p,j}}{G_{p,j+1}}\left[1 - \frac{\bar{p}_{j+1}}{\bar{z}_{j+1}}\frac{z_i}{p_i}\right]\right\}$$

or even further as

$$\left(\frac{\bar{p}}{\bar{z}}\right)_j = \frac{p_i}{z_i} - \frac{G_{p,j}}{G_{p,j+1}}\left(\frac{p_i}{z_i} - \frac{\bar{p}_{j+1}}{\bar{z}_{j+1}}\right) \quad \text{(C-3)}$$

Now, from pseudosteady-state theory,[15,16] the gas flow equation in terms of normalized pseudopressures is given by

$$\bar{p}_p - p_{pwf} = q_g\, b_{a,pss} \quad \text{(C-4)}$$

where

$$b_{a,pss} = 141.2\,\frac{\mu_{gi} B_{gi}}{k_g h}\left[\frac{1}{2}\ln\left(\frac{4}{e^{\gamma}}\frac{A}{C_A r_w^{\prime 2}}\right)\right] = \text{constant}$$

Eq. C-4 can also be written for two different points in time and solved for $b_{a,pss}$ in both cases. Since $b_{a,pss}$ is constant, the resulting relations may be equated to each other. Thus, for time levels j and $j+1$ it follows that

$$\frac{\bar{p}_{p,j+1} - p_{pwf,j+1}}{q_{g,j+1}} = \frac{\bar{p}_{p,j} - p_{pwf,j}}{q_{g,j}} \quad \text{(C-5)}$$

which may be rearranged as

$$\bar{p}_{p,j+1} = p_{pwf,j+1} + \frac{q_{g,j+1}}{q_{g,j}}(\bar{p}_{p,j} - p_{pwf,j}) \quad \text{(C-6)}$$

Eq. C-6, which is a flow type of equation, can be combined with Eq. C-3 to make it material-balance correct. The proposed method to couple these two equations in order to obtain the average pressure profile and gas in place volume is given below

Procedure

1. A table of p, z, p/z and p_p for the particular gas sample must be set up.
2. For the first time step (j =1), it is assumed that $\bar{p}_{p1} = \bar{p}_{pi}$. That is, the first average pseudopressure is assumed to be equal to the initial reservoir pseudopressure. Then, $\bar{p}_{p,j+1}$ can be recomputed from Eq. C-6 as

   $$\bar{p}_{p2} = p_{pwf2} + \frac{q_{g2}}{q_{g1}}(\bar{p}_{p1} - p_{pwf1})$$

   where again $\bar{p}_{p1} = \bar{p}_{pi}$.
3. Then, $\bar{p}_{j+1}$ and $\bar{z}_{j+1}$ ($\bar{p}_2$ and $\bar{z}_2$ for the first time step) are obtained using a look-up procedure in the table that was set up in step 1. Thus, $\frac{\bar{p}_{j+1}}{\bar{z}_{j+1}}$ can be estimated using the interpolated values.

4. Next, Eq. C-3 may be reentered with $\frac{\overline{p}_{j+1}}{\overline{z}_{j+1}}$ ($\frac{\overline{p}_2}{\overline{z}_2}$ for the first case) and $\frac{\overline{p}_j}{\overline{z}_j}$ ($\frac{\overline{p}_1}{\overline{z}_1}$ for the first case) can be back calculated as

$$\left(\frac{\overline{p}}{\overline{z}}\right)_j = \frac{p_i}{z_i} - \frac{G_{p,j}}{G_{p,j+1}}\left(\frac{p_i}{z_i} - \frac{\overline{p}_{j+1}}{\overline{z}_{j+1}}\right)$$

For the first time step, this calculation would be

$$\left(\frac{\overline{p}}{\overline{z}}\right)_1 = \frac{p_i}{z_i} - \frac{G_{p\,1}}{G_{p\,2}}\left(\frac{p_i}{z_i} - \frac{\overline{p}_2}{\overline{z}_2}\right)$$

Again, $\overline{p}_j$ and $\overline{p}_{p,j}$ are obtained using a look-up procedure entering the table made in step 1 with the recently calculated $\frac{\overline{p}_j}{\overline{z}_j}$.

5. Once more Eq. C-6 is used to recompute $\overline{p}_{p,j+1}$ with the last $\overline{p}_{p,j}$ found. For the first time step, $\overline{p}_{p2}$ would be recomputed using the last $\overline{p}_{p1}$.

6. Steps 2 through 4 must be repeated until no significant variation in two successive back substitutions for an average pressure at a given time step is obtained.

7. Once the right $\overline{p}_j$ is found ($\overline{p}_1$ for the first case), the same procedure is applied to obtain the values for the next time level.

It is worth noting that the back calculation proposed explains why Eq. C-3 is solved for the j time level, whereas C-6 is solved for the $j+1$ time level.

In order to ensure that the results obtained through this analysis technique are correct, a plot of gas in place, G, versus time may be made and a horizontal best fit trend should be constructed for G.

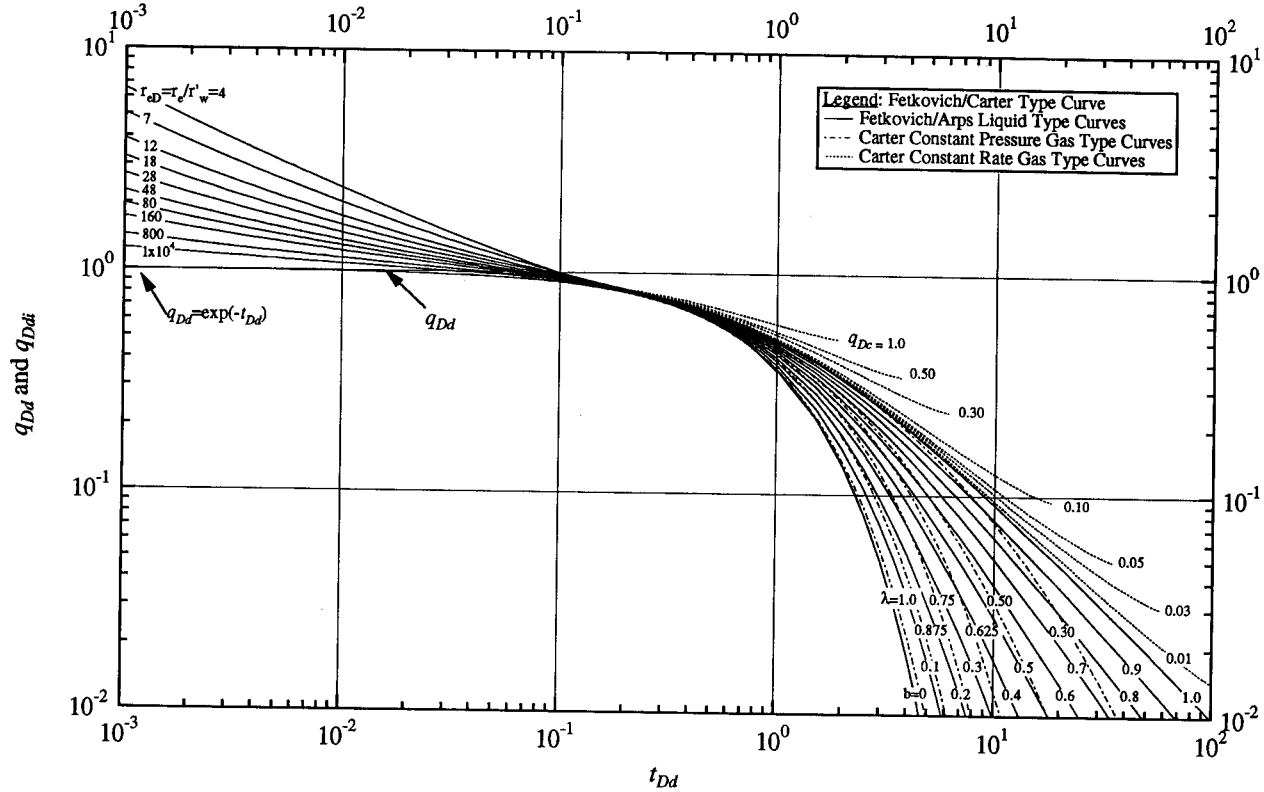

Figure 1 - Fetkovich Carter Type Curve.

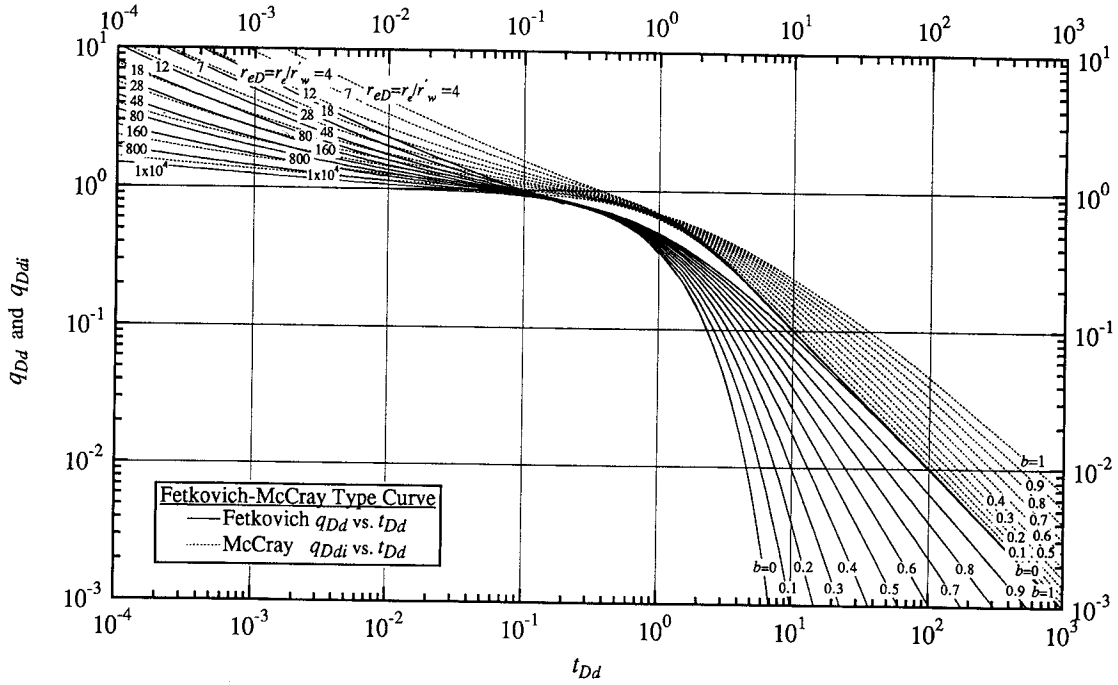

Figure 2 - Fetkovich-McCray Type Curve.

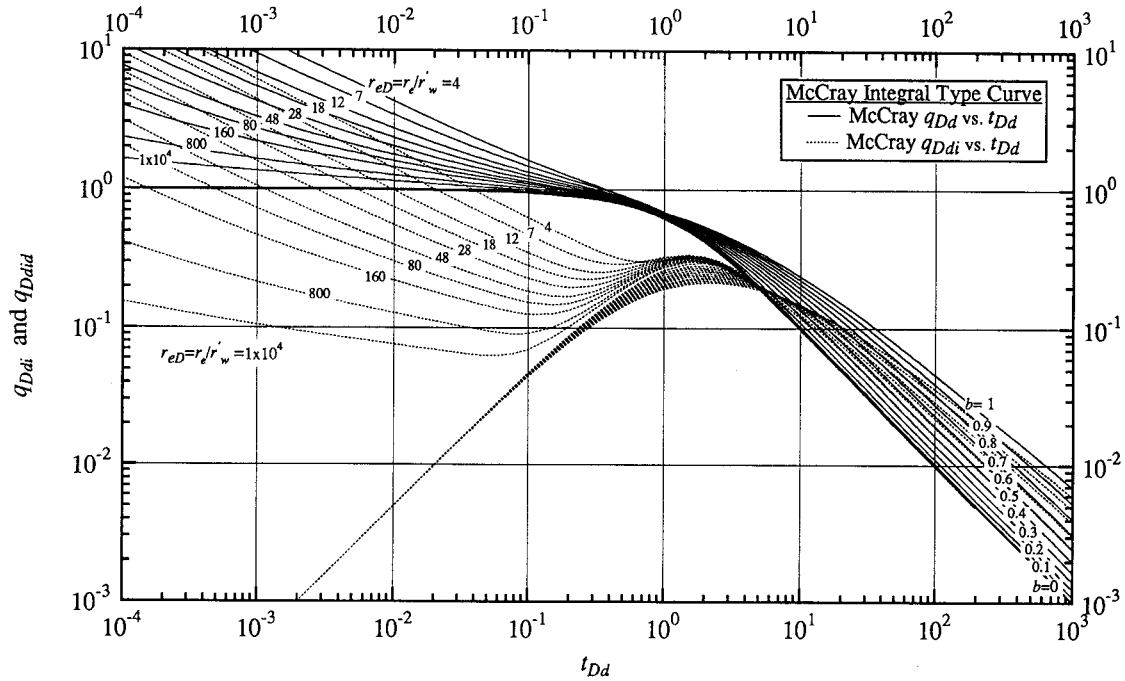

Figure 3 - McCray Integral Type Curve.

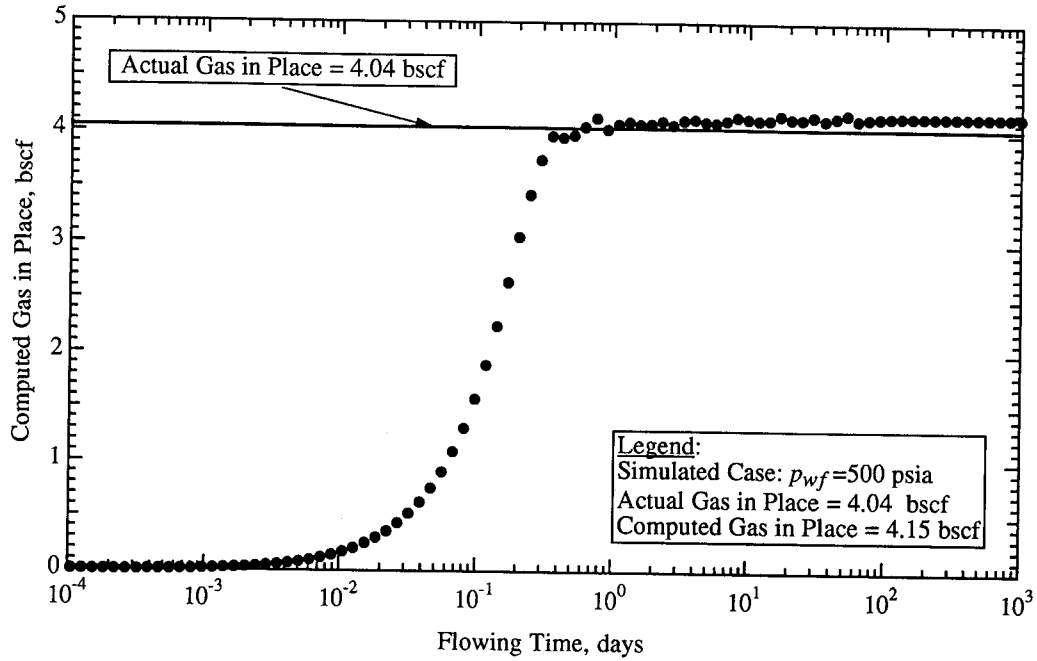

Figure 4 - Semilog Plot of Computed Gas in Place for a Simulated Example at Constant Bottomhole Flowing Pressure.

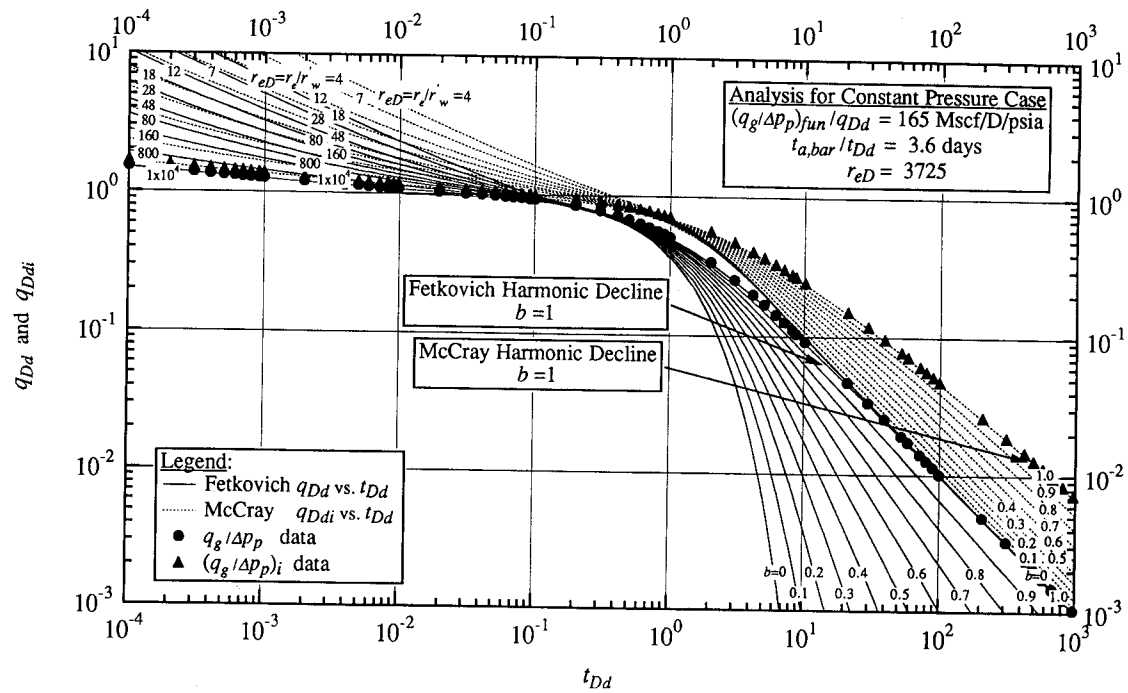

Figure 5 - Match of Normalized Flow Rate Functions on Fetkovich and McCray Type Curves for a Simulated Case of Constant Bottomhole Pressure.

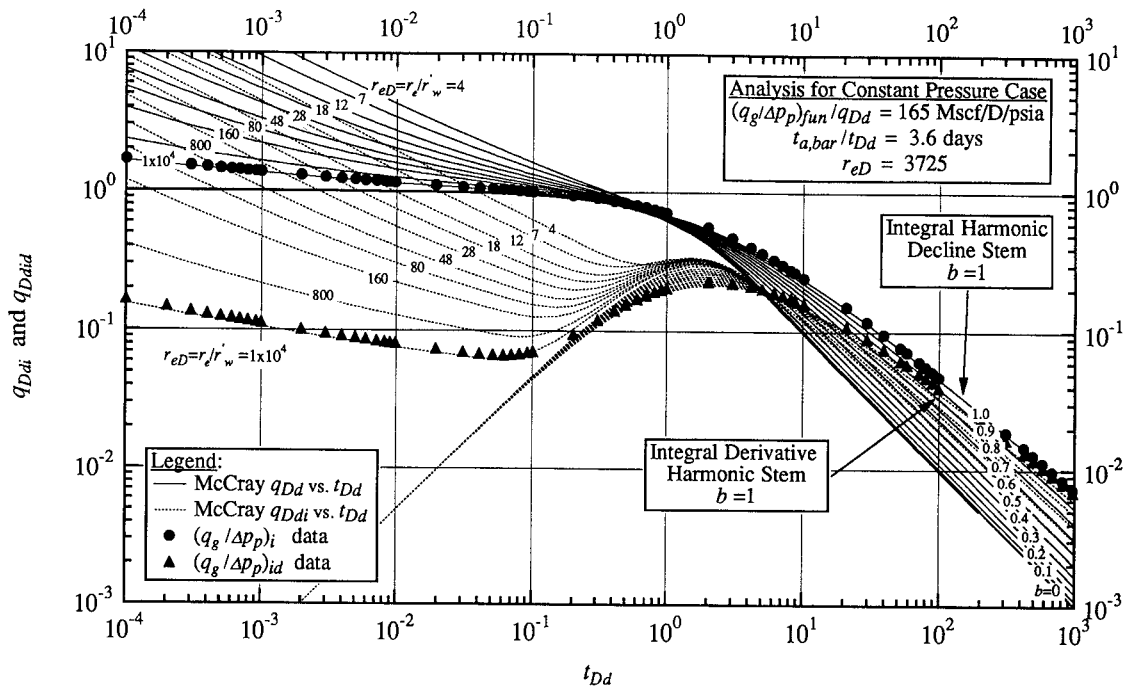

Figure 6 - Match of Normalized Flow Rate Integral and Integral Derivative on McCray Type Curve for a Simulated Case of Constant Bottomhole Pressure.

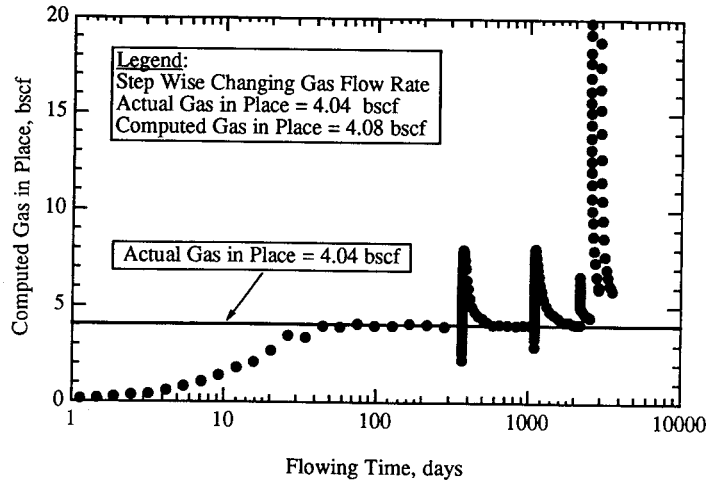

Figure 7 - Semilog Plot of Computed Gas in Place for a Simulated Example of Variable Flow Rate

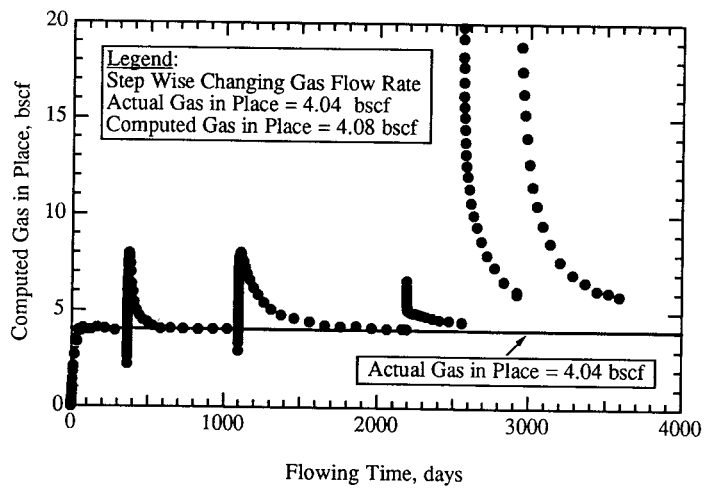

Figure 8 - Cartesian Plot of Computed Gas in Place for a Simulated Example of Variable Flow Rate

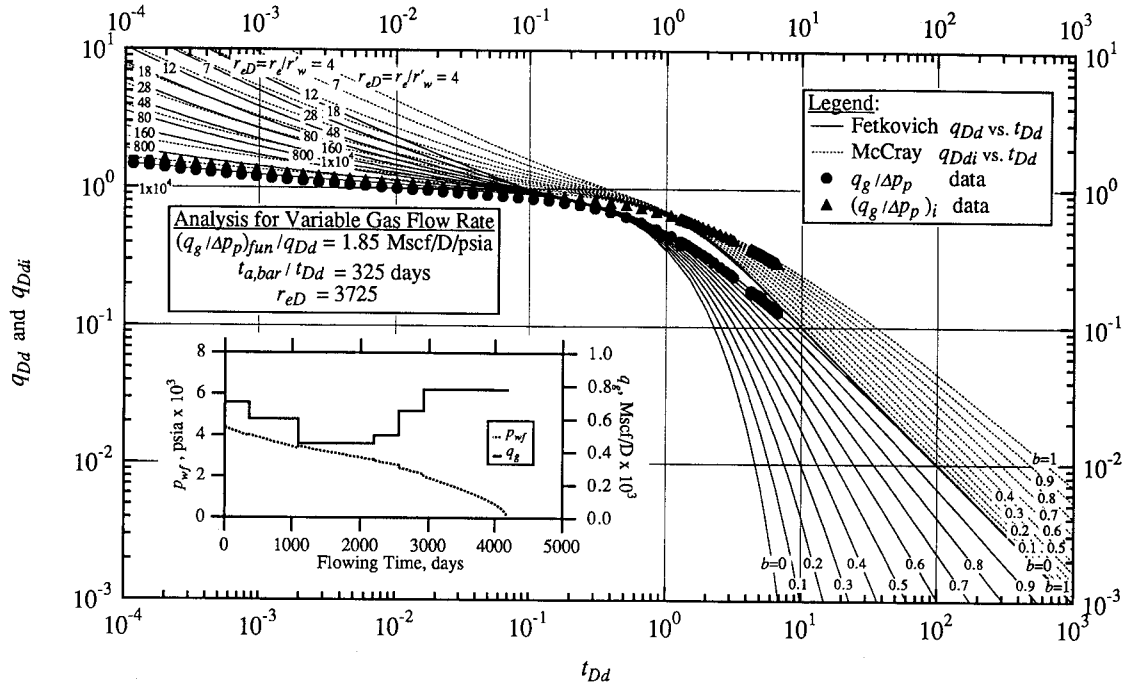

Figure 9 - Match of Normalized Flow Rate Functions on Fetkovich and McCray Type Curves for a Simulated Case of Variable Flow Rate.

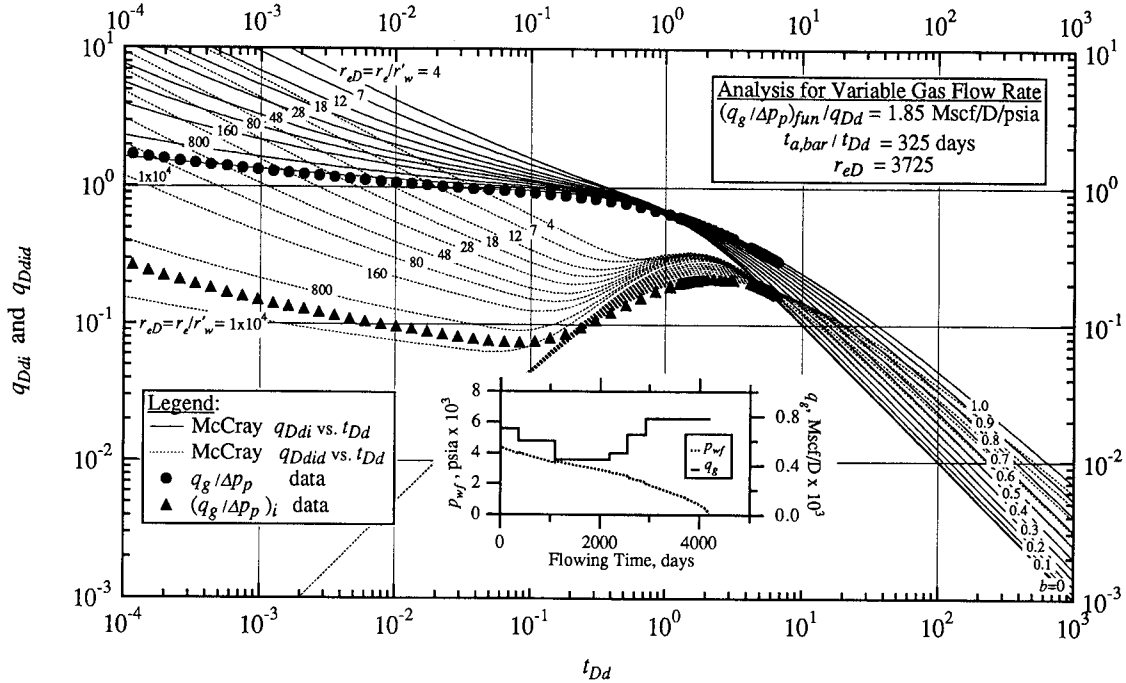

Figure 10- Match of Normalized Flow Rate Integral and Integral Derivative on McCray Type Curve for a Simulated Case of Variable Flow Rate.

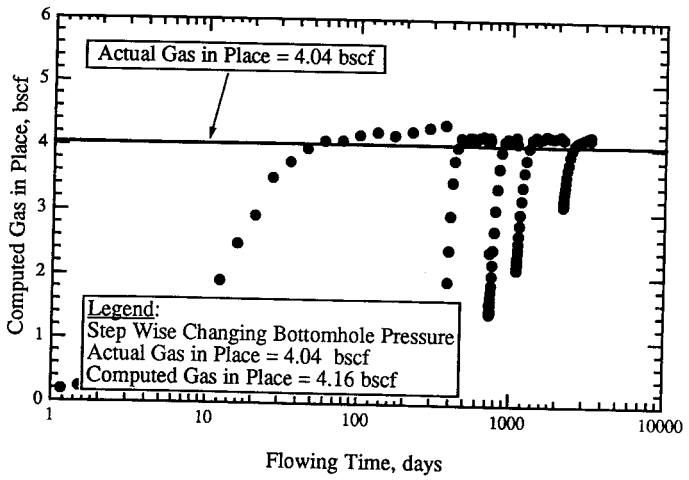

Figure 11 - Semilog Plot of Computed Gas in Place for a Simulated Example of Variable Bottomhole Pressure

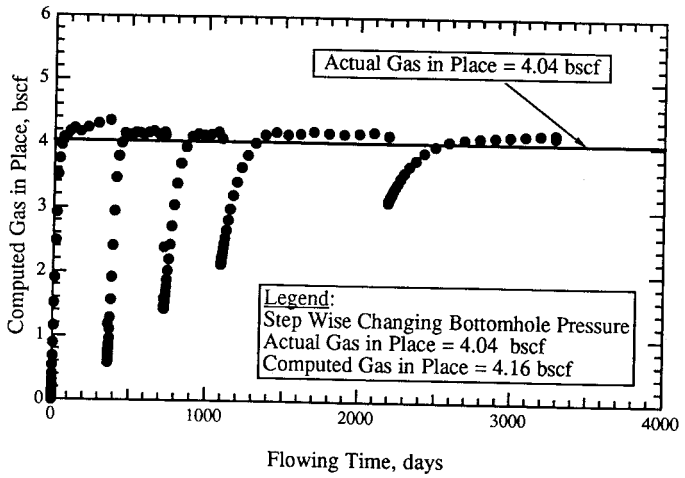

Figure 12 - Cartesian Plot of Computed Gas in Place for a Simulated Example of Variable Bottomhole Pressure

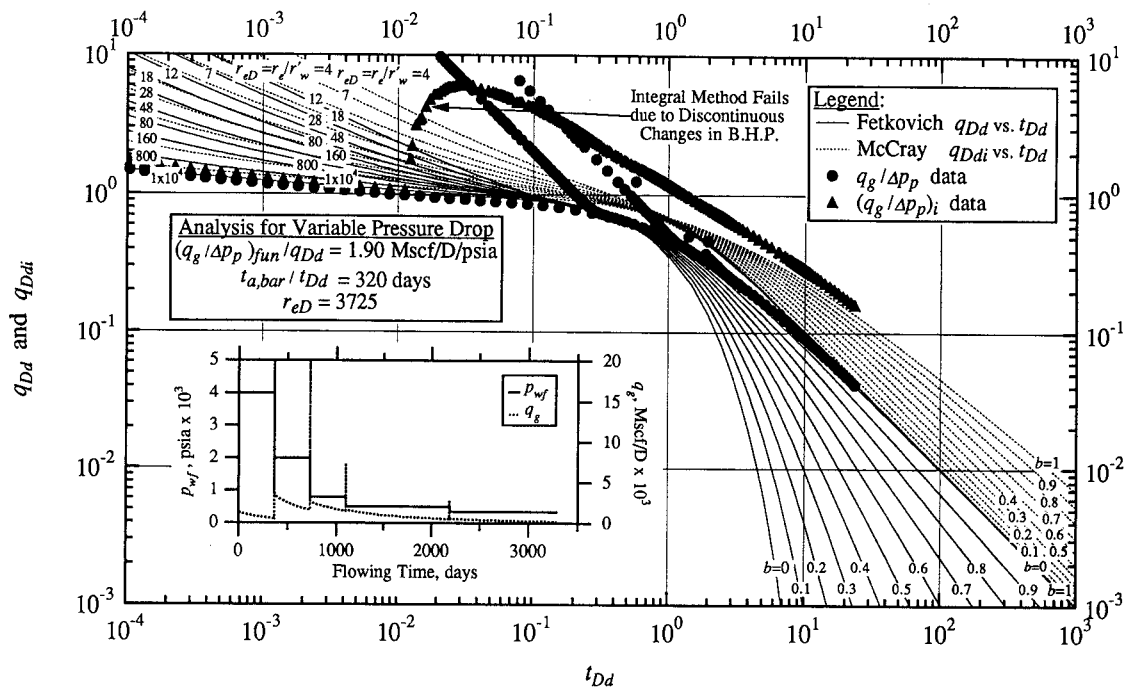

Figure 13 - Match of Normalized Flow Rate Functions on Fetkovich and McCray Type Curves for a Simulated Case of Variable Bottomhole Flowing Pressure.

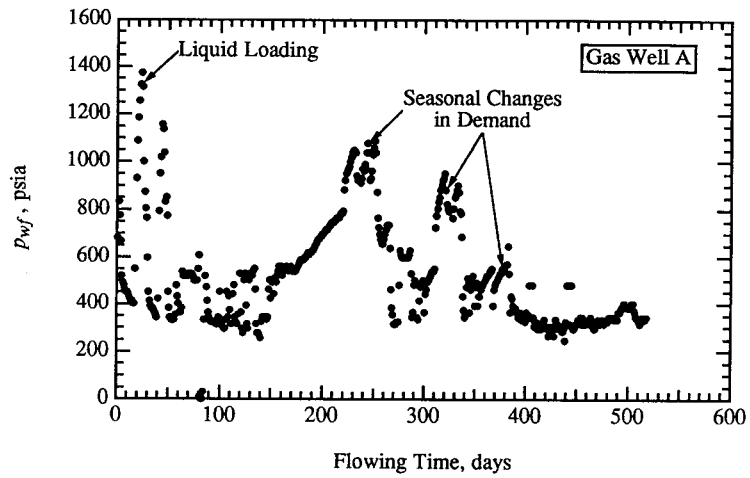

Figure 14 - Bottomhole Flowing Pressure Profile for Gas Well A.

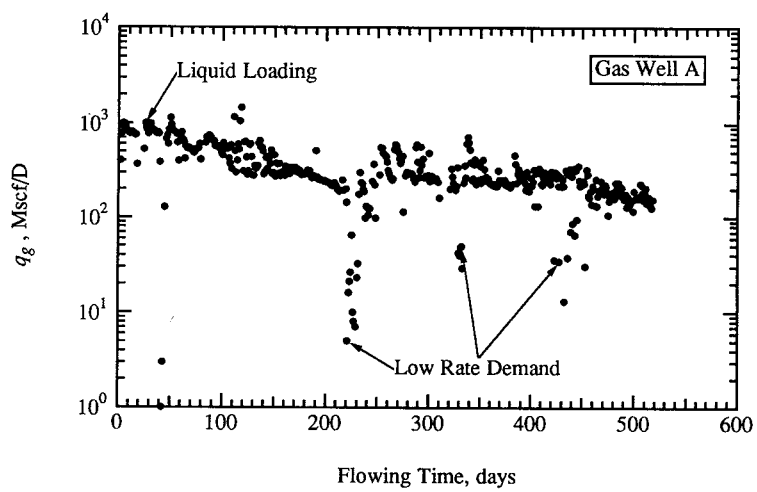

Figure 15 - Semilog Flow Rate Profile for Gas Well A.

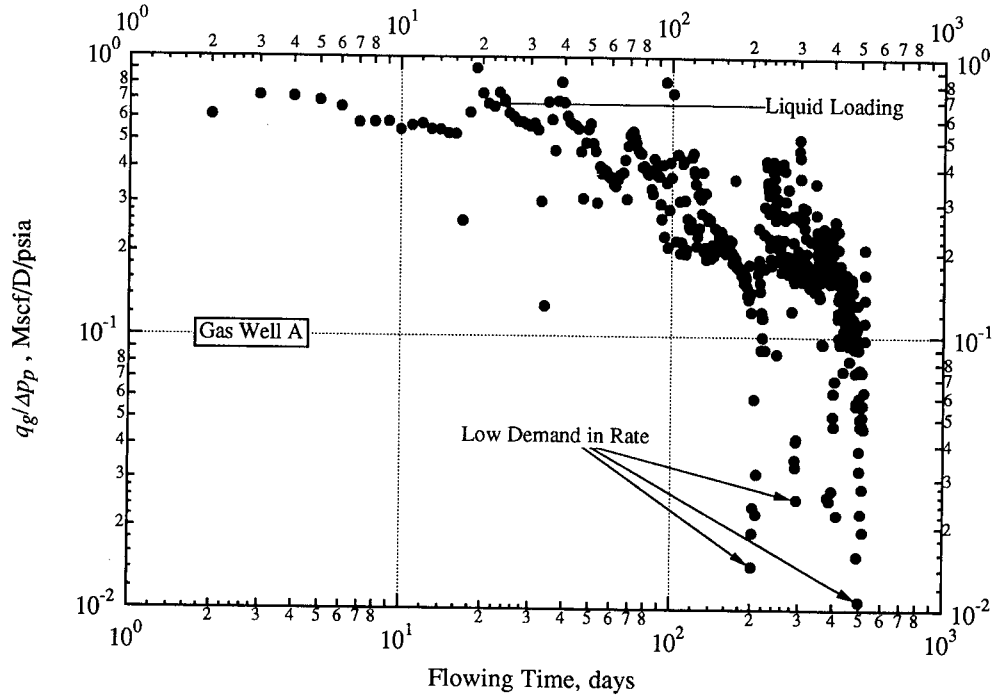

Figure 16 - Normalized Gas Flow Rate Profile in Time for Gas Well A.

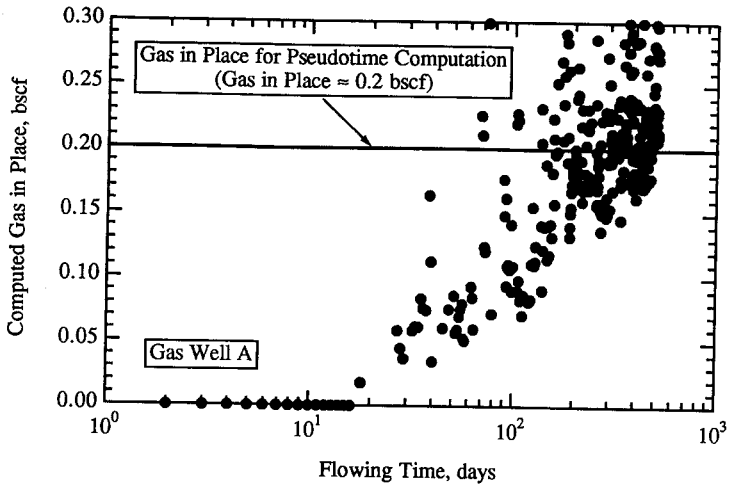

Figure 17 - Semilog Plot of Computed Gas in Place for Gas Well A.

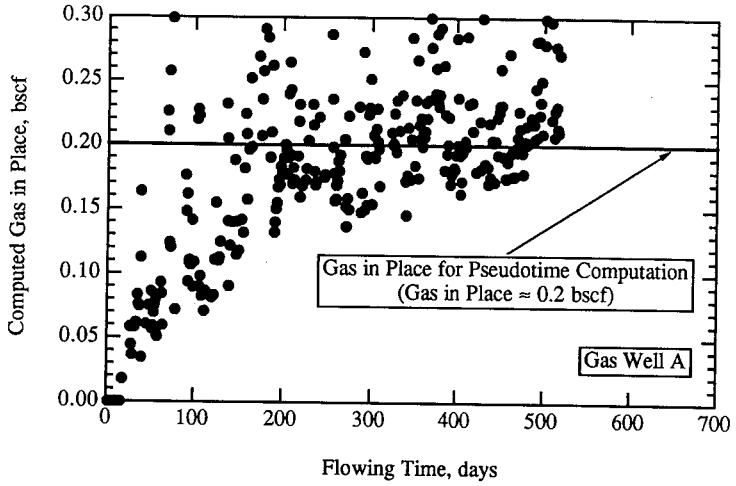

Figure 18 - Cartesian Plot of Computed Gas in Place for Gas Well A.

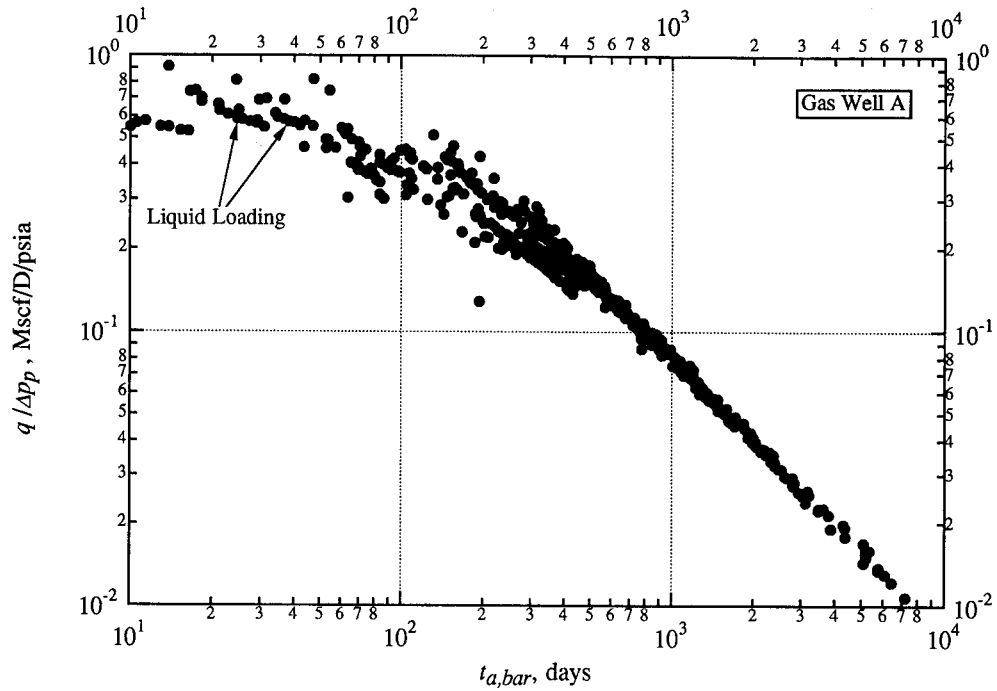

Figure 19 - Plot of Normalized Flow Rate versus Material Balance Pseudotime for Gas Well A.

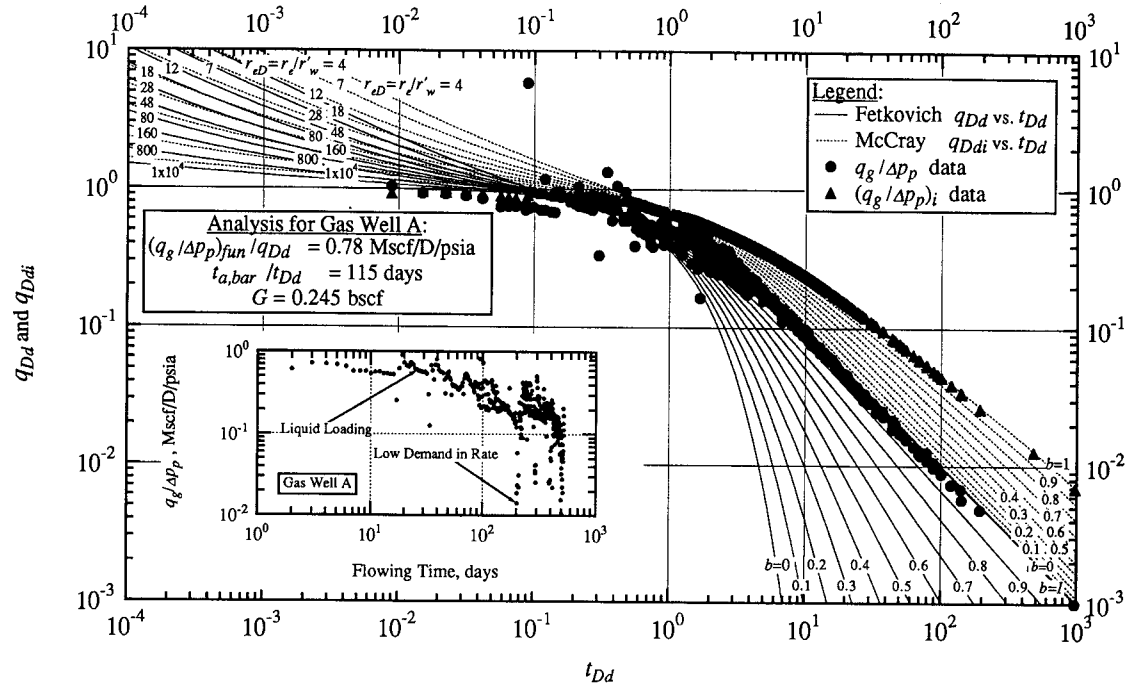

Figure 20 - Match of Normalized Flow Rate Functions on Fetkovich and McCray Type Curves for Gas Well A.

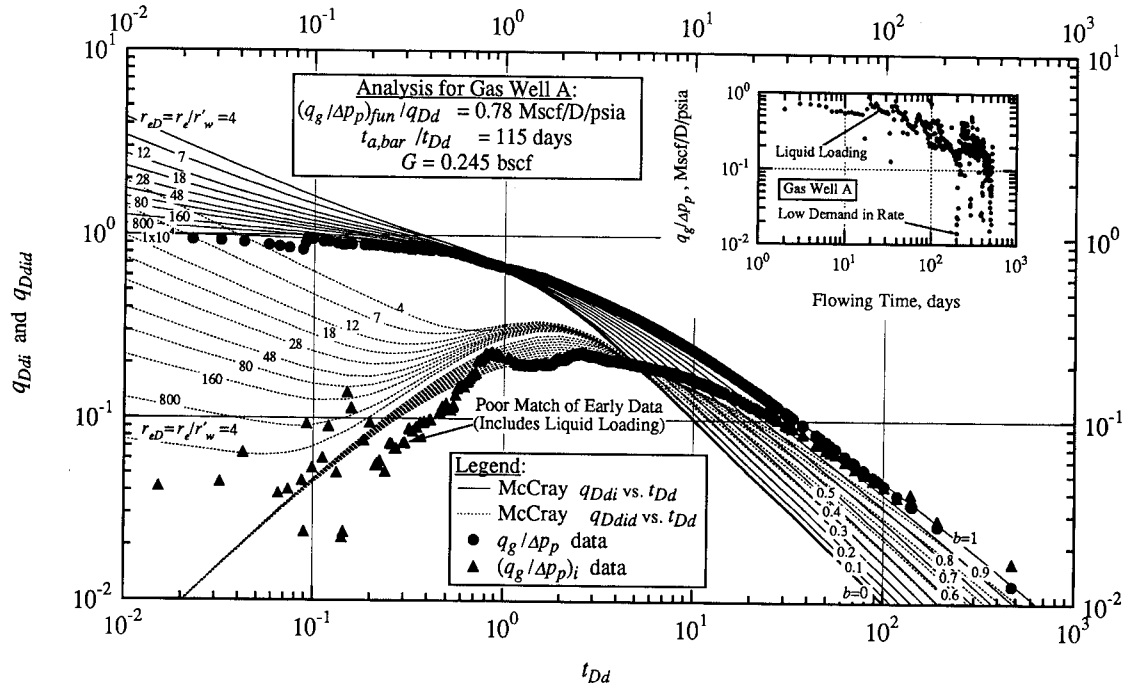

Figure 21 - Match of Normalized Flow Rate Integral and Integral Derivative on the McCray Type Curve for Gas Well A.